SEC P vs NP log 2026-04-28

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-12 04:30 UTC

SEC P vs NP — daily log 2026-04-28

32 cycles recorded

Block-Composition Sub-Additivity of Coarse Depth κ via the Roe-Trace Pairing

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Karchmer-Wigderson (CG-KW); Roe-style coarse cohomology / coarse geometry of metric spaces (geometric group theory + index theory branch of analysis) × Boolean formula depth lower bounds for KRW-style block compositions $f \circ g$, the central complexity-theoretic object behind $P \not\subset NC^1$; equivalently the depth complexity $D(f \circ g)$ of the composed function in the Karchmer-Raz-Wigderson program
- **Recorded**: 2026-04-28 01:39 UTC
- **Entry ID**: `8047f7f8eea1`

Statement

For all boolean functions $f:\{0,1\}^n \to \{0,1\}$ and $g:\{0,1\}^m \to \{0,1\}$ for which $HX^1(X_f)$ and $HX^1(X_g)$ admit nontrivial classes detected by Roe-controlled operators, the coarse depth κ defined via the trace_pairing on the KW-metric spaces obeys the sub-additive bound $\kappa(f \circ g) \geq \kappa(f) + \kappa(g) - C$, where C is a universal constant independent of n,m, and where $(X_{f \circ g}, d_{f \circ g})$ is built by the coarse_product operation. In particular, witness cocycles $c_f \boxtimes c_g \in HX^1(X_{f \circ g})$ and tensor Roe operators $T_f \otimes T_g \in C_u^*[X_{f \circ g}]$ realize $\log|\langle c_f \boxtimes c_g, T_f \otimes T_g\rangle| / \log(\text{propagation}(T_f \otimes T_g)) \geq \kappa(f) + \kappa(g) - C$.

Rationale

This sub-conjecture lifts axiom A2 (Composition sub-additivity for asdim) from the asymptotic-dimension layer to the strictly stronger κ layer, and chains through A1 (κ ≤ depth) to recover the KRW conjecture $D(f \circ g) \geq D(f) + D(g) - O(1)$. The mechanism is the multiplicativity of the Roe-index trace pairing on tensor products: $\langle c_f \boxtimes c_g, T_f \otimes T_g\rangle = \langle c_f, T_f\rangle \cdot \langle c_g, T_g\rangle$ (Connes-style product), while the propagation of $T_f \otimes T_g$ equals $\text{propagation}(T_f) \cdot \text{propagation}(T_g)$ under the depth-additive metric of coarse_product. Taking logs yields the additive bound. Should this hold even on small toy compositions, it provides direct empirical support for the KRW direct-sum theorem at the κ level. It does not contradict A3 (anti-natural-proofs) because random f has κ = O(1), so additivity holds trivially. It is consistent with A4 (anti-relativization) since coarse_product does not introduce oracle gates. A5 (Roe-index rigidity) underpins the stability of the tensor pairing under coarse equivalence.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 244, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 186, in run_trial
    kappa_f, kappa_g, kappa_fg = run_composition(f, g, k_f, k_g)
                                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 143, in run_composit
    X_f, d_f = build_X_d(f, k_f)
               ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 43, in build_X_d
    X = [tuple(i) for i in range(n)]
         ^^^^^^^^
TypeError: 'int' object is not iterable
```

Judge reasoning

The test crashed with a TypeError ('int' object is not iterable) at build_X_d before any κ values were computed, so no data exists to evaluate the pre-registered support criterion. | next: Fix build_X_d to enumerate vertices correctly (e.g., X = [tuple(b) for b in itertools.product([0,1], repeat=n)]) and rerun the 25-pair × 5-seed protocol.

Coboundary-Vanishing Threshold: HX^1 Triviality Below the Diameter-to-Depth Ratio for Parity-on-a-Tree

- **Verdict**: INCONCLUSIVE
- **Bridge**: Coarse Geometric Karchmer-Wigderson (CG-KW) framework / coarse (Roe) cohomology and geometric group theory applied to circuit complexity × Formula depth lower bounds for the recursive-parity (balanced AND/OR-tree of XORs) function in the NC^1 vs. AC^0[⊕] regime, expressed via the coarse 1-cohomology HX^1(X_f) and the Roe-trace pairing on C_u^*[X_f]
- **Recorded**: 2026-04-28 03:39 UTC
- **Entry ID**: 264c40c0d040

Statement

Let f_n = ⊕-tree_n be the balanced binary tree of XOR gates on n = 2^k inputs, with KW-metric space (X_{f_n}, d_{f_n}). Define the coboundary δ : C^0_coarse(X_{f_n}) -> C^1_coarse(X_{f_n}) and the propagation-R Roe subalgebra C_u^*[X_{f_n}]_R. Then there exists a universal constant α ∈ (0,1) such that, for all R ≤ α·log_2 n, every coarse 1-cocycle c with propagation ≤ R is a coboundary (so its class in HX^1 is trivial), while for R ≥ k = log_2 n there exist nontrivial classes c whose Roe-trace pairing satisfies log|⟨c,T⟩| / log(propagation(T)) ≥ k − O(1). Consequently κ(f_n) ≥ log_2 n − O(1), matching depth(f_n) = Θ(log n) up to constants and certifying axiom A1 sharply for an explicit recursive function.

Rationale

This conjecture directly tests axiom A1 (Coarse-KW link) together with A5 (Roe-index rigidity) on the canonical recursive function whose formula depth is exactly known (depth(⊕-tree_n) = log_2 n).

The KW protocol for the parity tree must descend the tree to find a differing leaf, so the controlled-cover multiplicity must grow by 1 per level; this should manifest as a vanishing-threshold for HX^1 at propagation R ≈ log_2 n, with cocycles at smaller propagation forced to be coboundaries. Verifying both the vanishing below the threshold and the nonvanishing above gives a sharp quantitative version of A1 on a function whose depth we can independently certify, providing a sanity check that κ is neither overshooting nor undershooting the true formula depth in a controlled recursive setting.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [2109.13292v1] The Hochschild Cohomology of Uniform Roe Algebras - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1906.11725v2] On the uniform Roe algebra as a Banach algebra and embeddings of ℓ_p uniform Roe algebras

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 133, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 73, in run_trial
    tree = xor_tree(n)
           ^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 20, in xor_tree
    left = xor_tree(n // 2)
           ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 22, in xor_tree
    return [left[i] ^ right[i] for i in range(n)]
            ~~~~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in the xor_tree construction before any data on R*(n), dim HX^1_R, or slope correlations was produced, so neither the support nor falsification conditions can be evaluated. | next: Fix the xor_tree recursion (the base case and/or input length handling causing the out-of-range index) and rerun the pre-registered protocol across n ∈ {4,8,16,32}.

Multiplicity-Doubling Lower Bound from Protocol Pullback on Indexing Gadgets

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Lifting (CGL) / Coarse Geometry & Roe-Algebra methods in Communication Complexity × Deterministic two-party communication complexity CC(f∘G^n) of lifted boolean functions
- **Recorded**: 2026-04-28 08:37 UTC
- **Entry ID**: `6a6a970e1f5b`

Statement

Let G = IND_b be the standard indexing MetricGadget on X = {0,1}^b × [b], Y = [b] with d the Hamming metric on X×Y, and let f: {0,1}^n → {0,1} have deterministic decision-tree complexity Q(f). Then for the lifted input space ((X×Y)^n, d_⊕) at scale R = diam(G), every deterministic protocol Π computing f∘G^n satisfies CoarsePullbackProtocol(Π) = (m_Π, R_Π) with m_Π ≥ 2^{Q(f)} / poly(b, n), so by axiom A2, CC(f∘G^n) ≥ Q(f) − O(log(bn)). Equivalently, CLC(f, IND_b) ≥ Q(f) · log_2(asdim_R(IND_b^{⊗n})+1) − O(log bn), recovering the Raz–McKenzie bound through the coarse-cover route.

Rationale

This sub-conjecture directly tests axiom A2 (Protocol → Cover) on the canonical gadget for which Raz–McKenzie lifting is known to hold, using axiom A3 (Asdim Amplification) to guarantee that the iterated indexing gadget has unbounded coarse dimension proportional to n. If A2 is sound, the multiplicity of the protocol-induced RoeCover must scale exponentially in the query complexity, otherwise one could collapse a query-tree lower bound through a low-multiplicity cover — contradicting the structure theorem at the heart of CGL. Confirming the bound on small parameters provides empirical evidence that the coarse-cover viewpoint reproduces the standard query-to-communication lift, a necessary sanity check before pursuing axiom A4's non-relativizing strengthening.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2211.17214v1] Lifting to Parity Decision Trees Via Stifling - [2105.01963v3] One-way communication complexity and non-adaptive decision trees - [2512.11833v1] Soft Decision Tree classifier: explainable and extendable PyTorch implementation - [1904.13056v4] Query-to-Communication Lifting Using Low-Discrepancy Gadgets - [2012.03415v1] On Query-to-Communication Lifting for Adversary Bounds

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38168056.py", line 114, in <module>
    seeds = [int(x) if x else 11 for x in input().split()]
                                          ^^^^^^^
EOFError: EOF when reading a line
```

Judge reasoning

The test crashed with an EOFError while reading seeds from stdin before any data could be produced, so neither the support nor falsification conditions can be evaluated. | next: Patch the harness to provide default seeds (or read from argv/env) when stdin is empty, then rerun the ≥20-trial battery over b∈{2,3,4}, n∈{2,3,4}.

Σ^b_0-PIND Width-2 Closure Round-Count vs DPLL Leaves

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Bounded arithmetic / Cook-Reckhow-Krajicek correspondence: the Σ^b_0-PIND round count r_2(F), defined as the minimum k ∈ ℕ ∪ {∞} such that iterating the closure operator Φ(S) = S ∪ {C : |C| ≤ 2 and C is RUP-derivable from F ∪ S in one unit-propagation pass} k times yields ⊥ ∈ Φ^k(∅); r_2 captures the depth of sharply-bounded polynomial-time

induction in PV restricted to width-2 reasoning (Cook-Reckhow lineage; rarely measured directly as a quantitative invariant on small unsatisfiable CNFs). × Lex-DPLL search-tree leaf count L(F) on small unsatisfiable 3-CNF formulas under a fixed lexicographic variable order with unit propagation, equivalently tree-resolution leaf count up to constants.
- **Recorded**: 2026-04-28 08:46 UTC
- **Entry ID**: `fc9124ecdaa8`

Statement

For every unsatisfiable 3-CNF F on $n \le 14$ variables and $m \le 6n$ clauses: (a) if $r\_2(F) < \infty$, then $\log_2 L(F) \le 2 \cdot r\_2(F) + 2 \cdot \log_2(n+1)$; (b) if $r\_2(F) = \infty$, then $\log_2 L(F) \ge n/4$. Equivalently, finiteness of width-2 RUP closure forces a polynomial DPLL bound, while non-closure forces an exponential one — a sharp dichotomy.

Rationale

Width-2 RUP closure is the propositional shadow of Cook's PV restricted to the sharply-bounded Σ^b_0-PIND fragment, the weakest layer of the Buss hierarchy. The Cook-Reckhow-Krajicek correspondence predicts a polynomial relation between sharply-bounded PV-proof depth and tree-like resolution size; quantifying it should expose the exact slack and yield a dichotomy mirroring the classical PHP/Tseitin obstructions to weak proof systems.

Novelty

- Judge: `NOVEL` over 11 arXiv hits

Top hits consulted: - [0209032v3] Complexity Results on DPLL and Resolution - [2407.17947v2] Supercritical Size-Width Tree-Like Resolution Trade-Offs for Graph Isomorphism - [1502.02131v2] Extracting verified decision procedures: DPLL and Resolution - [1411.7087v6] Consistency proof of a fragment of PV with substitution in bounded arithmetic - [2306.08535v1] Cyclic proofs for arithmetical inductive definitions

Empirical Test

- exit code: `124`, elapsed: `180.07s`

```
TIMEOUT after 180s
```

Judge reasoning

The test timed out after 180s (returncode=124) without producing any data, so neither the support nor falsification conditions could be evaluated. The critic also challenges, and no instances were verified. | next: Reduce the search space (e.g., cap n at 10 and reduce seeds per (n,α) pair) or parallelize/optimize the r_2 closure and DPLL leaf-count routines so the test completes within the time budget.

Mahler Measure of Truth-Table Polynomial Lower-Bounds DNF_min

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Number theory of polynomial heights — the (logarithmic) Mahler measure m(p) = log|a_d| + sum_{|α|>1} log|α| over the complex roots of p ∈ Z[x] (Lehmer-Boyd-Smyth tradition: Lehmer's problem on smallest non-cyclotomic measure, Boyd's heights, Smyth's resolved cubic case). Rarely deployed in meta-complexity; computable in poly time via numerical root-finding on the explicit degree-(2^n − 1) integer polynomial. × DNF_min(f): the minimum number of product terms in a DNF representation of a Boolean function f:{0,1}^n→{0,1} given its

2^n-bit truth table — the Hirahara–Oliveira–Santhanam restricted-MCSP proxy at the heart of meta-complexity worst-case-to-average-case reductions.
- **Recorded**: 2026-04-28 09:18 UTC
- **Entry ID**: `a6aec390c941`

Statement

Define the truth-table polynomial p_f(x) := sum_{i=0}$^{\{2}$n − 1} f(i)·x^i ∈ Z[x], where f(i) is the value of f on the n-bit binary expansion of i. For every f with |f^{-1}(1)| ≥ 1, m(p_f) ≤ (DNF_min(f) − 1) · log_2(n + 1). Equivalently, every f whose truth-table polynomial has logarithmic Mahler measure exceeding (k − 1)·log_2(n+1) satisfies DNF_min(f) > k.

Rationale

Each DNF product term corresponds to a coordinate-aligned subcube whose index set is a coset a+W ≤ F_2^n; its indicator polynomial factors as x^a · prod_j (1 + x$^{\{2}${i_j}}), so all roots lie on the unit circle and m = 0 (purely cyclotomic). Disjunction of k subcubes makes p_f a signed sum of k such cyclotomic-rooted polynomials, and Boyd-style sub-additive heuristics for Mahler measure under integer-coefficient sums predict at most a log_2(deg+1) escape per added term. If true, this transforms a hard Diophantine height into a poly-time-computable lower bound on DNF_min, giving a Lehmer-type obstruction useful in restricted-MCSP regimes.

Novelty

- Judge: “ over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by NATURAL_PROOFS barrier

Plünnecke-Ruzsa Doubling of Constraint Vectors Lower-Bounds SoS Degree for 3-XOR

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Additive combinatorics in F_2^n: the Plünnecke-Ruzsa doubling constant K(A)=|A+A|/|A| and its log version σ(A)=log_2|A+A|-log_2|A| of a finite set A⊆F_2^n (Freiman-Ruzsa / Bogolyubov / Sanders’ polynomial Bogolyubov tradition). The mathematical branch is harmonic / arithmetic combinatorics on abelian groups, distinct from polyhedral / Newton-polytope geometry, and rarely deployed in SoS lower bounds for CSPs. × Sum-of-Squares refutation degree d_SoS(F) of unsatisfiable 3-XOR instances F over GF(2), accessed via the Grigoriev-Schoenebeck combinatorial proxy g*(F) = min{|T| : ⊕*{i∈T} a_i = 0 ∈ F_2^n and* ⊕{i∈T} b_i = 1 ∈ F_2}, which sandwiches SoS degree to within ±1 by the Schoenebeck (2008) characterization for 3-XOR.
- **Recorded**: 2026-04-28 09:45 UTC
- **Entry ID**: `6ae7590ec4aa`

Statement

For every unsatisfiable 3-XOR instance F = ((a_i,b_i))_{i=1..m} on n ≤ 20 variables with constraint multiset A = {a_1,...,a_m} ⊆ F_2^n, the Schoenebeck combinatorial SoS-degree satisfies g*(F) ≥ σ(A) + 1, where σ(A) = log_2|A+A| - log_2|set(A)|. Equivalently, |A+A| ≤ |set(A)|·2^{g*(F)-1}.* A single (F, A) violating this inequality refutes the conjecture.

Rationale

Schoenebeck's theorem says small SoS degree forces a short additive dependency among constraint vectors; Plünnecke-Ruzsa says short additive dependencies push A toward an affine-subspace structure with low doubling. The bridge predicts a monotone, never-before-quantified link between structurelessness (large doubling) of the constraint vectors and the unavoidable length of a Schoenebeck certificate, transferring Freiman-Ruzsa intuition into proof complexity.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2507.01005v1] Discovery and study of the eclipsing variable star Grigoriev 1 - [0701009v2] Electron mobility on a surface of dielectric media: influence of surface level atoms - [1812.10216v2] Robust Approach for Rotor Mapping in Cardiac Tissue - [1701.04521v1] Sum of squares lower bounds for refuting any CSP - [0305179v3] Polynomial Degree and Lower Bounds in Quantum Complexity: Collision and Element Distinctness with Small Range

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_16090c09.py", line 92, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_16090c09.py", line 70, in run_trial
    set_A_size, A_plus_A_size, sigma_value = sigma(A)
                                             ^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_16090c09.py", line 56, in sigma
    A_plus_A.add(xor(a, b))
                 ^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_16090c09.py", line 21, in xor
    return x ^ y
           ~~^~~
TypeError: unsupported operand type(s) for ^: 'list' and 'list'
```

Judge reasoning

The test crashed with a TypeError in the xor() helper (xoring two lists), so no data was produced and the pre-registered support condition could not be evaluated. | next: Fix the xor() helper to operate on integer bitmask representations of F_2^n vectors (or convert lists via tuple/int) and re-run the 3000-instance sweep.

Kolmogorov Width of Lifted Functions Bounds MA^cc

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Descriptive complexity theory via Kolmogorov complexity × MA communication complexity
- **Recorded**: 2026-04-28 10:22 UTC
- **Entry ID**: `dd203ee122c9`

Statement

For any Boolean function f, the Kolmogorov width of the lifted function f ∘ G^n is at least Ω(log(n)) times the MA communication complexity of f, where G^n is the n-bit Indexing gadget.

Rationale

Kolmogorov complexity and descriptive complexity theory provide a framework for understanding the intrinsic complexity of objects, which can be applied to communication complexity to derive lower bounds. The use of lifting gadgets, such as the Indexing gadget, allows us to amplify the complexity of the function, making it more amenable to analysis.

Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [1811.04010v2] The layer complexity of Arthur-Merlin-like communication - [1904.13056v4] Query-to-Communication Lifting Using Low-Discrepancy Gadgets - [0506068v1] Quantum Arthur-Merlin Games - [1409.0584v2] Kolmogorov structure functions for automatic complexity - [1101.2946v3] Quantum Kolmogorov Complexity and Information-Disturbance Theorem

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9461c469.py", line 78, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9461c469.py", line 43, in run_trial
    lifted_f = [f[(i >> n) * (1 << n) + i % (1 << n)] for i in range(2**(n+n))]
                ~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError before producing any data, so neither the support nor falsification conditions could be evaluated. No empirical evidence exists to assess the conjecture. | next: Fix the indexing bug in the lifted function construction (the formula `(i >> n) * (1 << n) + i % (1 << n)` is malformed for the Indexing gadget) and rerun the 120 trials.

Tensor-Amplified Asdim Forces Linear Communication Blow-up on Inner-Product Gadgets

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Lifting (CGL) / Coarse Geometry & Asymptotic Dimension (geometric group theory applied to communication complexity) × Deterministic two-party communication complexity of lifted functions f∘G^n, with G obtained by k-fold GadgetComposition of a base inner-product gadget IP_2

- **Recorded**: 2026-04-28 10:39 UTC
- **Entry ID**: `916caa07c776`

Statement

Let IP_2 be the 2-bit inner-product MetricGadget on X=Y={0,1}^2 with Hamming metric d, and let G_k = IP_2^{⊗k} under coordinatewise additive metric. For every Boolean f:{0,1}^n→{0,1} with deterministic query complexity Q(f), the CoarseLiftingComplexity satisfies CLC(f, G_k) ≥ Q(f) · log_2(k+1) − O(n), and consequently CC(f∘G_k^n) ≥ Q(f)·log_2(k+1) − O(n). Equivalently, AsdimCertify(LiftedInputSpace(G_k), R=diam(G_k)) returns multiplicity m ≥ k+1, so by axiom A2 every deterministic protocol Π must have cost ≥ log_2(k+1) per query bit recovered.

Rationale

This sub-conjecture directly tests axiom A3 (Asdim Amplification) chained with A2 (Protocol→Cover). A3 predicts that iteratively tensoring IP_2 (which already has asdim ≥ 1 because its 16-point input space contains a metric line of length 4) yields G_k with asdim ≥ k − O(1). A2 then converts the cover multiplicity into a communication lower bound. If the conjecture holds, it provides a concrete constructive route to lifting bounds that scale with the tensor depth k, recovering Raz–McKenzie–style log-factor blow-ups from a purely coarse-geometric witness rather than from gadget-specific combinatorics. Failure would falsify A3 in the small-multiplicity regime and force a refinement of how asdim composes under ⊗.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1904.13056v4] Query-to-Communication Lifting Using Low-Discrepancy Gadgets - [1104.4468v3] A strong direct product theorem for quantum query complexity - [2511.17227v3] A Lifting Theorem for Hybrid Classical-Quantum Communication Complexity - [0111062v2] One-way communication complexity and the Neciporuk lower bound on formula size - [1110.6876v4] Lower bounds for topological complexity

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_af1c65ed.py", line 185, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_af1c65ed.py", line 124, in run_trial
    G_k = gadget_composition([ip_2] * k)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_af1c65ed.py", line 74, in gadget_compos
    result[i*n+k][j*n+l] = gadgets[i][k] + gadgets[j][l]
                           ~~~~~~~~~~^^^
TypeError: 'function' object is not subscriptable
```

Judge reasoning

The test crashed with a TypeError in gadget_composition (treating a function as subscriptable) before any data was produced, so neither the support nor falsification criteria could be evaluated. | next: Fix the gadget_composition implementation to operate on metric matrices/arrays rather than function objects, then re-run the 12 (k,n,f) instances across 3 seeds.

Möbius Defect of NW Design Lattice Bounds Parity Bias

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Möbius theory of finite lattices (Rota–Stanley combinatorial Möbius inversion on intersection lattices; Stanley EC1 Ch.3) — rarely deployed in PRG bias analysis × Output bias of the Nisan–Wigderson pseudorandom generator NW_D$^{f:\{0,1\}}$d→{0,1}^m against linear (parity) distinguishers χ_T, T⊆[m]
- **Recorded**: 2026-04-28 10:48 UTC
- **Entry ID**: `b27f028c321c`

Statement

Let D=(S_1,…,S_m) be an NW design over universe [d] with |S_i|=n and pairwise intersections ≤k. Let L(D) be the meet-semilattice of all distinct intersections T_I=⋂*{i∈I} S_i (I⊆[m]) ordered by ⊆ with bottom 0̂=∅, and let μ_L be its Möbius function. Define the Möbius defect δ(D)=Σ*{T∈L(D), T≠∅} |μ_L(0̂,T)|·2^{−|T|}. Conjecture: for every f:{0,1}^n→{0,1} with max Walsh coefficient |f̂(U)|≤2^{−n/3} on |U|≥k, and every non-empty T⊆[m], |E_{y∼U_d}[χ_T(NW_D^f(y))]| ≤ 4·δ(D).

Rationale

Standard NW bias bounds use the crude factor m·2·ε from a hybrid argument that ignores higher-order inclusion–exclusion cancellation across the design's intersection pattern. Möbius values μ_L(0̂,T) are precisely the signed multiplicities by which intersections of design rows are over/under-counted, so replacing m by δ(D) should yield substantially tighter bias estimates whenever L(D) is structured (e.g., projective-plane or polynomial designs versus random subset-systems). If the conjecture holds, δ(D) becomes a computable, design-only certificate of fooling power independent of f beyond the Fourier-tail hypothesis.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2601.03730v1] Perception-Aware Bias Detection for Query Suggestions - [1207.0393v2] Pseudo-finite hard instances for a student-teacher game with a Nisan-Wigderson generator - [2109.14047v1] Second Order WinoBias (SoWinoBias) Test Set for Latent Gender Bias Detection in Coreference Resolution - [0710.4339v1] Heavy-Quark Masses from the Fermilab Method in Three-Flavor Lattice QCD - [1111.0981v1] Form factors for B to Kll semileptonic decay from three-flavor lattice QCD

Empirical Test

- exit code: `0`, elapsed: `0.03s`


```
{"TRIAL": null}
{"TRIAL": null}
{"TRIAL": null}
{"TRIAL": null}
{"TRIAL": null}
```


Judge reasoning

The test stdout contains only `{"TRIAL": null}` entries with no measurements, no RESULT line, and no evidence that the pre-registered criteria (≥200 designs, both constructions, regression slope ≥0.5, zero violations) were evaluated. The critic also flags a likely vacuous-bound saturation issue, so SUPPORTED cannot be claimed. | next: Fix the test harness so trials actually emit (D,f,T,bias,δ(D)) tuples, then verify whether 4·δ(D) is non-trivial (<1) for the tested regime before re-running the p

Sandpile-Group Order of Certificate Conflict Graph Lower-Bounds Q^dt

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Abelian sandpile models / chip-firing on finite graphs (Bjorner–Lovász–Shor; Holroyd–Levine–Mészáros–Peres–Propp–Wilson); the sandpile (critical) group K(G) computed as the cokernel of the reduced graph Laplacian, equivalently the spanning-tree count by Kirchhoff's matrix-tree theorem × Decision-tree depth Q^dt(f) of small Boolean functions f:{0,1}^k→{0,1}, transported via the Raz–McKenzie / Göös–Pitassi–Watson Indexing-gadget lifting theorem to deterministic two-party communication complexity D^cc(f∘IND_b)
- **Recorded**: 2026-04-28 11:19 UTC
- **Entry ID**: `ef886a7094f0`

Statement

For every Boolean f:{0,1}^k→{0,1}, let G_f be the graph whose vertices are the minimal 1-certificates of f and whose edges {C,C'} connect certificates that fix some common variable to opposite values (call these 'conflicts'). Then log_2 |K(G_f)| ≤ Q^dt(f) · log_2(k+1), where |K(G_f)| is the order of the abelian sandpile group of G_f, which equals the number of spanning trees of G_f by Kirchhoff (defined to be 1 if G_f has ≤1 vertex or is disconnected with each component a tree). By GPW lifting, this entails D^cc(f∘IND_b) ≥ Ω(log_2 |K(G_f)| · log b / log k).

Rationale

The minimal 1-certificate clutter captures the irreducible branching obligations any decision tree must witness; their conflict graph encodes exactly the pairs that cannot coexist on a single root-to-leaf path, so its cycle-richness should constrain protocol/decision-tree branching budgets. The sandpile group is a torsion abelian invariant whose log-size is Kirchhoff's spanning-tree count — a single integer determinant — and a depth-d tree induces at most k^d distinct query histories, plausibly bounding how much such cyclic redundancy can be 'covered'. Chip-firing and sandpile invariants have essentially never been deployed in lifting-style lower bounds, giving an algebraic-combinatorial obstruction distinct from rank, Dilworth-width, Möbius, or asymptotic-dimension routes.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [0906.2809v2] Sandpile groups and spanning trees of directed line graphs - [2602.01042v2] On Condensation of Block Sensitivity, Certificate Complexity and the AND (and OR) Decision Tree Co - [1904.12209v2] Sandpile monomorphisms and limits - [2512.11833v1] Soft Decision Tree classifier: explainable and extendable PyTorch implementation - [0811.2800v2] Parallel Chip-Firing on the Complete Graph: Devil's Staircase and Poincare Rotation Number

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6f6dc07d.py", line 131, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6f6dc07d.py", line 104, in run_trial
    if not is_minimal_1_certificate(f, C):
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6f6dc07d.py", line 25, in is_minimal_1_
    if f[C[:i] + (1 - C[i],) + C[i+1:]] == f[C]:
          ~~~~~~^~~~~~~~~~~~~
TypeError: can only concatenate list (not "tuple") to list
```

Judge reasoning

The test crashed with a TypeError (list/tuple concatenation bug in is_minimal_1_certificate) before any of the 800 trials produced data, so the pre-registered support condition cannot be evaluated. | next: Fix the type bug by representing certificates consistently (e.g., always tuples) so that C[:i] + (1 - C[i],) + C[i+1:] type-checks, then rerun the 800-trial sweep.

Stanley-Reisner Projective Dimension Lower-Bounds Monotone-KW Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial commutative algebra: projective dimension pd(R_Δ) of the Stanley-Reisner ring R_Δ = GF(2)[x_1,...,x_n]/I_Δ accessed via Hochster's formula β_{i,T}(R_Δ) = dim tilde H_{|T|-i-1}(Δ|_T; GF(2)), in the Stanley–Hochster–Reisner–Terai tradition (Miller–Sturmfels, Combinatorial Commutative Algebra). Rarely deployed against circuit/formula complexity; distinct from polytope/Newton-polytope or matroid-rank approaches. × Monotone Karchmer–Wigderson game depth D_m(f) of monotone Boolean functions f:{0,1}^n→{0,1}, equivalently monotone formula depth in the {AND,OR}-basis via the Karchmer–Wigderson equivalence — the central object of Razborov-style monotone NC^1 lower bounds.
- **Recorded**: 2026-04-28 11:47 UTC
- **Entry ID**: `e180d6571db0`

Statement

Define Δ_f := {S ⊆ [n] : f(1_S)=0}, the simplicial complex of non-1-certificates of a monotone f, and let Pd(f) := max{|T|−j−1 : T ⊆ [n], j ≥ −1, tilde H_j(Δ_f|_T; GF(2)) ≠ 0} (with Pd(f)=0 if all reduced homologies vanish). Then for every monotone f, the monotone-KW protocol depth satisfies D_m(f) ≥ Pd(f). One counterexample (a monotone f with D_m(f) < Pd(f)) refutes the conjecture.

Rationale

A monotone-KW protocol decomposes the rectangle MIN(f)×MAX(¬f) into coordinate-labeled monochromatic rectangles; this rectangular decomposition is the combinatorial shadow of a free resolution of I_{Δ_f}, whose length is exactly pd(R_{Δ_f}) by Auslander–Buchsbaum. Hochster's formula reduces pd to vanishing of reduced GF(2)-homology on induced subcomplexes, an invariant that is rarely linked to KW games (prior monotone bounds use fooling sets, lattice rank, or Razborov approximators). Because the conjecture targets monotone classes only, it sidesteps the natural-proofs barrier (which constrains P/poly with arbitrary basis under crypto assumptions, not monotone NC^1).

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [1801.08709v2] Adaptive Lower Bound for Testing Monotonicity on the Line - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1101.4831v1] The f–vector of the clique complex of chordal graphs and Betti numbers of edge ideals of uniform hypergraphs

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_850c8ff1.py", line 96, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_850c8ff1.py", line 75, in run_trial
    D_m_f = compute_D_m(f, n)
            ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_850c8ff1.py", line 22, in compute_D_m
    min_f = [f(i) for i in range(2**n)]
             ^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_850c8ff1.py", line 19, in <lambda>
    return lambda s: all(x in s for x in range(n))
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_850c8ff1.py", line 19, in <genexpr>
    return lambda s: all(x in s for x in range(n))
                         ^^^^^^
TypeError: argument of type 'int' is not iterable
```

Judge reasoning

The test crashed with a TypeError in the f-construction lambda before any instance was evaluated, so no data was produced and the pre-registered support condition was not met. | next: Fix the buggy monotone-f constructor (the lambda treats integer states as iterable) and rerun the full 20+168+500 enumeration.

Lie-Stabilizer Dimension of Clause Cubic Bounds DPLL Leaves

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometric Complexity Theory: dimension of the stabilizer Lie algebra g_p = {A ∈ Mat_n(Q) : Σ_i (A·x)_i ∂_i p(x) ≡ 0 as a polynomial} of a homogeneous polynomial p, computed by Gaussian elimination on the linear system extracted from the cubic-monomial coefficients; the canonical first-order Mulmuley–Sohoni / Bürgisser–Ikenmeyer obstruction used to separate det_n from perm_n via stabilizer dimension ($2n^2-1$ vs. ~2n−2). Distinct from apolar/catalecticant rank and from Hessian rank. × Lex-DPLL search-tree leaf count L_DPLL(F) on small UNSAT 3-CNF formulas at clause density α, equivalently tree-resolution leaf count up to constants — the canonical proof-complexity proxy for refutation hardness.
- **Recorded**: 2026-04-28 12:19 UTC
- **Entry ID**: `77253411c65d`

Statement

Define p_F ∈ Q[x_1,...,x_n] by p_F(x) = Σ_{clauses C} ε_C · x_{i_1}x_{i_2}x_{i_3}, where ε_C = (−1)^{#negated literals in C} and (i_1,i_2,i_3) are the indices of C's variables; let g_F be its GL_n stabilizer Lie algebra and dim_Q(g_F) its dimension. Conjecture: for every UNSAT 3-CNF F on n ≥ 8 variables, log_2 L_DPLL(F) + dim_Q(g_F) ≤ n + 2·⌈log_2 n⌉. A single UNSAT F violating this inequality (with positive margin) refutes it.

Rationale

GCT identifies high stabilizer dimension with maximally structured polynomials (the determinant's huge symmetry is exactly what trivializes its computation), and the orbit-closure containment program reads lower bounds off symmetry-budget mismatches. Translated to refutations, an UNSAT formula whose clause cubic carries large Lie symmetry should expose unit-propagation/decomposition structure, while symmetry-poor cubics correspond to the 'rigid' regime where DPLL is forced to branch combinatorially. Cubic clause polynomials are the cheapest GCT-style invariant capturing 3-SAT, and their stabilizer Lie algebras have not, to my knowledge, been linked to proof complexity.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1212.5380v2] On properties of principal elements of Frobenius Lie algebras - [2411.11095v3] Invariant theory and coefficient algebras of Lie algebras - [1401.3795v2] Relationship between Nichols braided Lie algebras and Nichols algebras - [1611.00827v2] Geometric complexity theory and matrix powering - [0911.3482v5] Complexity of Networks (reprise)

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92d0288f.py", line 139, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92d0288f.py", line 74, in run_trial
    M = [[0] * (n**2) for _ in range(binomial(n+2, 3))]
                                 ^^^^^^^^^^^^^^^^^^^^^^
TypeError: 'float' object cannot be interpreted as an integer
```

Judge reasoning

The test crashed with a TypeError (binomial returned a float passed to range) before any trials were run, so no Δ(F) values were produced and neither the support nor falsification condition can be evaluated. | next: Fix the binomial call by casting to int (or use math.comb) and rerun the 1000-instance sweep to obtain actual Δ(F) statistics.

Følner-Defect Floor Forces Logarithmic Communication Overhead on Equality-Lifted Gadgets

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Lifting (CGL) — coarse geometry / geometric group theory applied to communication complexity × Deterministic two-party communication complexity CC(EQ_k ∘ G^n) of the lifted Equality function on n copies of a fixed MetricGadget G
- **Recorded**: 2026-04-28 12:42 UTC
- **Entry ID**: `723804a64233`

Statement

Let G = (X,Y,g,d) be a MetricGadget with |X|=|Y|=k and Hamming-graph metric d, and let EQ_k : {0,1}^n → {0,1} denote Equality on n bits with Q(EQ_k) = n. Define the Følner defect of the lifted space at scale R as δ_R(n) = min over finite F ⊂ (X×Y)^n of |∂_R F|/|F|, where *∂R F is the d*⊕-R-boundary. Then for every gadget G satisfying axiom A4 (Property A) and every n, CC(EQ_k ∘ G^n) ≥ Q(EQ_k) · ⌈log_2(1/δ_R(n) + 1)⌉ for R = diam(G), and equivalently CLC(EQ_k, G) is bounded below by n · log_2(AsdimCertify(LiftedInputSpace_n, diam(G))+1). In particular, when AsdimCertify returns multiplicity m at scale R, every deterministic protocol must use at least n · log_2(m+1) bits.

Rationale

This sub-conjecture directly tests Axiom A2 (Protocol → Cover) combined with Axiom A4 (Property-A Transfer). A2 says any cost-c protocol yields a RoeCover of multiplicity ≤ 2^c, so log_2 of the minimum cover multiplicity lower-bounds CC. A4 promotes that bound by the query complexity factor Q(f). For Equality, Q(EQ_k) = n and any non-trivial RoeCover at scale diam(G) of the n-fold lift must have multiplicity at least k^Ω(1) per coordinate by amenability obstructions (Følner defect), giving a per-coordinate logarithmic multiplier. The conjecture should hold because the Følner-defect quantity δ_R(n) shrinks geometrically in n for product metrics, forcing the cover multiplicity — and hence the protocol cost — to scale linearly with n with a logarithmic-in-k coefficient.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [0111062v2] One-way communication complexity and the Neciporuk lower bound on formula size - [9906008v2] A Lower Bound on the Average-Case Complexity of Shellsort - [1904.13056v4] Query-to-Communication Lifting Using Low-Discrepancy Gadgets - [1411.4413v2] Observation of the rare $B_s^0 \to \mu^+\mu^-$ decay from the combined analysis of CMS and LHCb data - [0901.0512v4] Expected Performance of the ATLAS Experiment - Detector, Trigger and Physics

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d88ab7be.py", line 147, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d88ab7be.py", line 123, in run_trial
    cc_eqk_gn_value = cc_eqk_gn(G, n)
                      ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d88ab7be.py", line 88, in cc_eqk_gn
    C = [[0 for _ in range(m)] for _ in range(m)]
                                        ^^^^^^^^

TypeError: 'float' object cannot be interpreted as an integer
```

Judge reasoning

The test crashed with a TypeError before any data was produced (‘float’ object cannot be interpreted as an integer in cc_eqk_gn at line 88), so no configurations were evaluated and the pre-registered support criterion cannot be assessed. | next: Cast m to int (e.g., m = int(round(m))) when constructing the cost matrix in cc_eqk_gn, then rerun the 12 configurations to obtain CC vs n·log_2(m_n+1) data.

Bipartite Token-Sliding Diameter Lower-Bounds Monotone-KW Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial reconfiguration / token-sliding graphs (Ito-Demaine-Hearn-Etsuko-Uno tradition): the diameter D_TS(B_f) of the token-sliding reconfiguration graph on the bipartite minterm/maxterm incidence structure of a monotone Boolean function f, where tokens sit on minterms (1-side) or maxterms (0-side) and a slide moves a token along a ‘shared coordinate’ edge. × Monotone Karchmer-Wigderson game depth D_m(f) for monotone Boolean functions, equivalently monotone formula depth in the {AND,OR} basis
- **Recorded**: 2026-04-28 12:48 UTC

- **Entry ID**: `299d5162a4f6`

Statement

For every monotone Boolean function f on n variables, the monotone-KW depth satisfies D_m(f) >= log2(D_TS(B_f) + 1) - 1, where D_TS(B_f) is the diameter (in the connected component containing the canonical all-coordinates token configuration) of the token-sliding graph on the bipartite minterm-maxterm incidence B_f under the move 'slide token from minterm m to minterm m' iff m and m' differ in exactly one coordinate i with f(m')=1, and similarly on the maxterm side'. Equivalently, no monotone formula of depth d can resolve every (minterm, maxterm) pair within fewer than 2^(d+1)-1 token slides, so depth d < log2(D_TS+1)-1 is forbidden for every f. One reconfiguration-diameter witness D_TS(B_f) > 2^(d+1) refutes any claimed monotone formula of depth <= d for f.

Rationale

A monotone-KW protocol of depth d defines a labeling of (minterm, maxterm) pairs by coordinates that respects f-monotonicity; concatenating coordinate-labels along any reconfiguration path yields a protocol-consistent path on the incidence bipartite graph, and protocol depth bounds the number of distinct decisions reachable by any single token-slide trajectory. Reconfiguration diameters are well-studied combinatorially but, to our knowledge, have never been pulled back through the KW correspondence as a depth lower-bound certificate, despite providing a direct combinatorial obstruction (long sliding paths force deep protocols). The bridge isolates a non-spectral, non-monotone-circuit-counting invariant that is robust under taking sub/superfunctions and sidesteps the natural-proofs barrier because D_TS is not closed under random monotone projections.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [2504.10671v2] Token Sliding Reconfiguration on DAGs - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [1904.06696v3] On bipartite distance-regular Cayley graphs with diameter 3 - [2302.03695v4] The Harer-Zagier and Jackson formulas and new results for one-face bipartite maps

Empirical Test

- exit code: `0`, elapsed: `0.03s`


```
{"TRIAL": {"seed": 11, "metric_name": "monotone-KW depth", "metric_value": null, "instances_tested":
{"TRIAL": {"seed": 23, "metric_name": "monotone-KW depth", "metric_value": null, "instances_tested":
{"TRIAL": {"seed": 37, "metric_name": "monotone-KW depth", "metric_value": null, "instances_tested":
{"TRIAL": {"seed": 53, "metric_name": "monotone-KW depth", "metric_value": null, "instances_tested":
{"TRIAL": {"seed": 71, "metric_name": "monotone-KW depth", "metric_value": null, "instances_tested":
```


Judge reasoning

All 5 trials returned metric_value=null, instances_tested=0, and counterexample='mapping_undefined', meaning the test never computed D_m(f) or D_TS(B_f) for any function. No empirical evidence was produced, and the pre-registered threshold of >=25 functions was not met. | next: Disambiguate the token-sliding move rule (especially the maxterm-side clause) and implement a concrete D_TS computation plus a monotone-KW depth oracle for small n (3..5) before scaling to n=8.

Magnus Level-2 Defect Bounds DNF_min via Truth-Table Inversions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Magnus expansion / Chen's iterated-integral signature in noncommutative formal power series of free groups (Magnus 1935; Chen 1957; Reutenauer, Free Lie Algebras 1993; Lyons rough-path tradition). The level-2 'inversion defect' λ_2(w) = |#{i<j : w_i=0, w_j=1} − #{i<j : w_i=1, w_j=0}| is exactly the pairing of the truncated Magnus signature S^{≤2}(w) ∈ Q⟨⟨A,B⟩⟩ with the commutator [A,B], a free-Lie-algebra invariant computable in one O(|w|) pass and rarely deployed in meta-complexity. × DNF_min(f), the minimum number of product terms in a DNF representation of a Boolean function f:{0,1}^n→{0,1} presented by its 2^n-bit truth table — the Hirahara–Oliveira–Santhanam restricted-MCSP proxy used in meta-complexity worst-case-to-average-case reductions and the canonical truth-table-input meta-complexity object.
- **Recorded**: 2026-04-28 13:27 UTC
- **Entry ID**: `8bf8c5c64dba`

Statement

Let T = TT(f) ∈ {0,1}$^{\{2}$n} be the lex-ordered truth table of f and define λ_2(f) = |asc(T) − desc(T)| where asc(T) = #{i<j : T_i=0, T_j=1} and desc(T) = #{i<j : T_i=1, T_j=0}. Then for every Boolean f on n variables, log_2(1 + λ_2(f)) ≤ n + log_2(1 + DNF_min(f)) + 5·log_2(n+1). A single f violating this inequality refutes the conjecture; for uniformly random f one has log_2 λ_2(f) ≈ 1.5 n by anti-concentration, so the bound is exponentially below random and forces DNF_min(f) = 2^{Ω(n)} only on the trivial counting fraction.

Rationale

The Magnus expansion linearizes a binary word into a free Lie algebra; its level-2 [A,B]-coefficient is the inversion-pair imbalance of the truth-table string, computable in a single O(2^n) sweep. Low-DNF functions have truth tables decomposing as the union of M structured sub-rectangles whose individual inversion imbalances align in a near-cancelling way, so λ_2 should grow like M·2^n rather than the random 2^{1.5n}. The hardness predicate 'λ_2(f) ≥ 2^{n+log M+O(log n)}' is satisfied by a 1−exp(−Ω(n)) fraction of random f only as a trivial counting bound, and the small-λ_2 side (which is the structural side actually used to upper-bound DNF_min) holds for an exponentially tiny fraction of random f, dodging the Razborov–Rudich largeness condition and hence the natural-proofs barrier.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [0810.5488v1] The Magnus expansion and some of its applications - [1501.01331v1] DNF complexity of complete boolean functions - [1407.6169v3] Multiplicative Complexity of Vector Valued Boolean Functions - [1509.08963v3] Integrals of Lipschitz-Hankel Type, Legendre Functions, and Table Errata - [2402.02935v4] Nuclear mass table in deformed relativistic Hartree-Bogoliubov theory in continuum, II: Even-Z nuclei

Empirical Test

- exit code: `0`, elapsed: `146.77s`


```
{"TRIAL": {"seed": 11, "metric_name": "lambda_2 vs DNF_min", "metric_value": 7.0276395303248895, "ins
{"TRIAL": {"seed": 23, "metric_name": "lambda_2 vs DNF_min", "metric_value": 7.00788077049084, "insta
{"TRIAL": {"seed": 37, "metric_name": "lambda_2 vs DNF_min", "metric_value": 7.00585013217493, "insta
{"TRIAL": {"seed": 53, "metric_name": "lambda_2 vs DNF_min", "metric_value": 7.041945775829029, "inst
{"TRIAL": {"seed": 71, "metric_name": "lambda_2 vs DNF_min", "metric_value": 7.017263172398402, "inst
RESULT: SUPPORTED mean=7.020115876243618 std=0.013375328575307719 support_fraction=1.0
```

Judge reasoning

The test reports SUPPORTED with only 16 instances per seed at unspecified n, falling far short of the pre-registered ≥840 instances across n∈{8,…,20} and four ensembles; the critic's saturation argument (log_2(1+λ_2) ≤ 2n makes the bound vacuous until n≥28) means the tested regime cannot probe potential failures. The pre-registered support condition is not unambiguously met. | next: Run the full pre-registered protocol at n∈{8,10,12,14,16,18,20} with ≥30 seeds per (n,ensemble,M) cell, explicit

---

Monomer-Dimer Entropy Lower-Bounds Tseitin Resolution Width

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Statistical mechanics on graphs / monomer-dimer theory: Heilmann-Lieb real-rootedness of the matching polynomial m_G(x)=Σ_k (-1)^k m_k(G) x^{n−2k} and the resulting per-site free energy h_md(G)=(1/n) log Σ_k m_k(G), a Heilmann-Lieb / Godsil / Gurvits invariant rarely deployed in proof complexity and distinct from spectral expansion. × Resolution refutation width w(T(G,c)⊢⊥) of Tseitin formulas on 3-regular graphs G with odd total F_2-charge c — the Urquhart / Ben-Sasson–Wigderson / Alekhnovich–Razborov target.
- **Recorded**: 2026-04-28 13:54 UTC
- **Entry ID**: `c5c4b106a0cf`

Statement

There exists an absolute constant α > 0 such that for every 3-regular graph G on n vertices and every charge c ∈ F_2^{V(G)} with Σ_v c_v = 1, the resolution refutation width satisfies w(T(G,c)⊢⊥) ≥ α · n · h_md(G), where h_md(G) := (1/n) · log_2 Σ_{k=0}^{⌊n/2⌋} m_k(G). A single 3-regular graph on at most 20 vertices with odd charge whose Tseitin formula admits a resolution refutation of width strictly below 0.05 · n · h_md(G) refutes the conjecture.

Rationale

Heilmann-Lieb log-concavity makes h_md a topological-combinatorial 'routing entropy' of unit F_2-flows in the cycle space, and these routings are exactly the obstruction narrow resolution must clear to cancel an odd parity. Unlike edge expansion, h_md is girth-sensitive and survives moderate bottlenecks, so the conjecture predicts width bounds strictly stronger than Ben-Sasson–Wigderson on low-expansion-but-high-h_md graphs such as long prisms or generalized Möbius-Kantor families. The tie to extended Frege is downstream: if the bound saturates the BSW expansion bound only on Ramanujan-type graphs, the slack pinpoints where higher proof systems can compress.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations - [0407622v3] A Study of the Composition of Ultra High Energy Cosmic Rays Using the High Resolution Fly's Eye - [2209.05839v3] On bounded depth proofs for Tseitin formulas on the grid; revisited - [1809.06500v3] Range entropy: A bridge between signal complexity and self-similarity - [2309.13632v2] Decay of correlations in the monomer-dimer model

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_31fe51cd.py", line 189, in <module>
```

```
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_31fe51cd.py", line 159, in run_trial
    h_md_value = h_md(G)
                 ^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_31fe51cd.py", line 52, in h_md
    u, v = e
    ^^^^
ValueError: too many values to unpack (expected 2)
```

Judge reasoning

The test crashed with a ValueError in the h_md function (edge unpacking failure) before producing any data, so no instances were evaluated against the pre-registered criterion. | next: Fix the edge iteration in h_md (handle multi-edges/self-loops or unpack edges correctly) and rerun the experiment over the specified 3-regular graphs and odd charges.

Sprague-Grundy Game Value of CNF Bounds Tree-Frege Lines

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Sprague-Grundy / Nim-value theory of impartial combinatorial games (Bouton 1901; Sprague 1935; Grundy 1939; Berlekamp-Conway-Guy 'Winning Ways' tradition); a branch of combinatorial game theory rarely deployed in proof complexity, distinct from Karchmer-Wigderson games and from Ehrenfeucht-Fraisse pebble games used in descriptive complexity. × Tree-like Frege refutation line count L_TF(F) on small unsatisfiable 3-CNFs, accessed via Reckhow's polynomial equivalence with tree-resolution leaf count L_T(F⊢⊥) (depth-0 Frege over CNFs), serving as a tractable proxy for the Krajicek-Buss-Razborov-Impagliazzo EF lower-bound program.
- **Recorded**: 2026-04-28 14:25 UTC
- **Entry ID**: `501b089a21b9`

Statement

Define the impartial Variable-Splitting Game V(F) on a CNF F as follows: a state is a CNF obtained from F by a sequence of literal substitutions with clause simplification (drop satisfied clauses, drop falsified literals from surviving clauses); a move from state F' picks any variable v occurring in F' together with a value b∈{0,1} and transitions to F'[v=b]; terminal states (no remaining variables) carry Sprague-Grundy value 0, and non-terminal values are the mex of children's values. Let G(F) denote this Grundy value. Conjecture: for every unsatisfiable 3-CNF F with n ≤ 20 variables, ⌊log_2(1+G(F))⌋ ≤ ⌈log_2 L_T(F⊢⊥)⌉, where L_T is the leaf count of lex-DPLL with unit propagation; a single unsatisfiable 3-CNF violating this inequality refutes the conjecture.

Rationale

Sprague-Grundy values capture an exponential structural complexity intrinsic to the move-graph of CNF splitting, yet have never been linked to proof-system line counts; if the inequality holds, G(F) yields a Frege-style certificate of refutation hardness computable without exhibiting any refutation, opening a non-Boolean-Fourier route from impartial-game algebra to EF size lower bounds via Reckhow's reduction. The Grundy value is invariant under literal-flipping and variable-renaming, matching a symmetry class compatible with natural-proofs-safe invariants. Failure modes (small G(F) with large L_T) are equally informative, exposing a genuine gap between game-theoretic disjunctive complexity and proof-theoretic refutation length.

Novelty

- Judge: `NOVEL` over 7 arXiv hits

Top hits consulted: - [1912.12573v3] A base-p Sprague-Grundy type theorem for p-calm subtraction games: Welter's game and representations of generalized symm - [1702.07068v1] The Sprague-Grundy function for some nearly disjunctive sums of Nim and Silver Dollar games - [1507.04665v2] Invariance of the Sprague-Grundy Function for Variants of Wythoff's Game - [2207.14140v1] Playing a 2D Game Indefinitely using NEAT and Reinforcement Learning - [1104.3098v3] Optimal strategies for a game on amenable semigroups

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ad3b71d3.py", line 107, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ad3b71d3.py", line 83, in run_trial
    F = generate_cnf(n, alpha)
        ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ad3b71d3.py", line 66, in generate_cnf
    v1, v2 = random.sample(variables, 2)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 413, in sample
    raise TypeError("Population must be a sequence.  "
TypeError: Population must be a sequence.  For dicts or sets, use sorted(d).
```

Judge reasoning

The test crashed with a TypeError in generate_cnf (random.sample called on a non-sequence) before any instances were evaluated, so neither the violation count nor the Pearson correlation was measured. | next: Fix generate_cnf to pass a list (e.g., random.sample(list(variables), 2)) and rerun the pre-registered 300-instance sweep.

Roe-Skeleton Spectral Gap Forces Multiplicity in Protocol-Induced Covers on Small-Diameter Gadget Lifts

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Lifting (CGL) — Coarse Geometry / Roe-Algebra Operator Theory applied to Lifting in Communication Complexity × Deterministic two-party communication complexity of f∘G^n, specifically the multiplicity profile m_Π of the ProtocolPullback partition on the LiftedInputSpace (X×Y)^n
- **Recorded**: 2026-04-28 14:45 UTC
- **Entry ID**: `a510fd856726`

Statement

Let G = (X, Y, g, d) be a MetricGadget with R := diam(G) and let A_R denote the 0/1 adjacency matrix of the RoeAlgebraSkeleton at scale R on (X×Y)^n under the d_⊕ metric, with combinatorial Laplacian L_R = D_R − A_R. Let λ_R(n) be the smallest nonzero eigenvalue of L_R, and let N = |X×Y|^n. Then for every deterministic communication protocol Π computing any non-constant f∘G^n, the ProtocolPullback multiplicity satisfies m_Π · R_Π ≥ ⌈ λ_R(n) · N / (4 · max-degree(A_R)) ⌉, and consequently the CoarseLiftingComplexity invariant obeys CLC(f, G) ≥ log_2(λ_R(n) · N /

(4·deg)) for every f with D(f) ≥ 1. Equivalently, a vanishing Roe-skeleton spectral gap is a coarse obstruction that lower-bounds protocol cover multiplicity, in alignment with axiom A2 (Protocol → Cover) and A4 (Property-A Transfer).

Rationale

This concretizes axiom A2 by predicting a quantitative spectral certificate: the Roe algebra skeleton's Cheeger-type gap controls the minimum multiplicity any transcript-induced cover must exhibit at scale R. The Cheeger inequality on the skeleton graph implies the boundary-to-bulk ratio of any low-diameter set is at least λ_R/(2·deg); since a transcript pullback of scale R_Π and multiplicity m_Π yields a cover where each cell has d_⊕-diameter ≤ R_Π, the spectral gap forces the cover cells to be 'fat' — i.e., to overlap. A protocol with multiplicity 1 (a partition) at scale R is precluded whenever λ_R > 0, certifying that asdim ≥ 1 in the lifted geometry. This also gives a lattice on which Property-A (axiom A4) can be cross-checked, since amenable/Property-A spaces have shrinking gaps but persistent multiplicity, providing a concrete bridge between coarse spectral data and communication lower bounds.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1903.06084v8] Lifting coarse homotopies - [2312.08907v2] Roe algebras of coarse spaces via coarse geometric modules - [2408.07701v2] A categorical interpretation of Morita equivalence for dynamical von Neumann algebras - [2403.19550v2] Spectral gap for surfaces of infinite volume with negative curvature - [1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c7ce6686.py", line 178, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c7ce6686.py", line 119, in run_trial
    for gadget, X, Y, d, R in [(X_XOR, Y_XOR, d_XOR, R_XOR), (X_IND, Y_IND, d_IND, R_IND)]:
        ^^^^^^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 5, got 4)
```

Judge reasoning

The test crashed with a ValueError (tuple unpacking mismatch) before producing any data, so neither the support nor the falsification condition can be evaluated. | next: Fix the iterable in run_trial to include the gadget label (5-tuple) and rerun the deterministic test to obtain λ_R(n), N, and max-degree values.

Walsh-Code Antichain Width of NW Designs Bounds PRG Linear Bias

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Extremal set theory: LYM/Dilworth antichain width of the design's intersection poset, scored by Walsh-character weight (a Sperner-LYM invariant rarely used in PRG analysis, distinct from Möbius/lattice or expander arguments) × Output linear (parity) bias of the Nisan-Wigderson generator NW_{D,f}: {0,1}^d -> {0,1}^m against parity distinguishers chi_T, T subset of [m], built from a hard function f:{0,1}^l -> {0,1}

- **Recorded**: 2026-04-28 14:54 UTC
- **Entry ID**: `52d3382c4243`

Statement

Let D=(S_1,...,S_m) be an (l,a)-NW design over [d] with a<=log2(m), and let P_D be the poset on nonempty subsets of [m] ordered by T<=T' iff the symmetric-difference indicator of (union of S_i for i in T) is a 2-adic refinement of that of T'. Define the Walsh antichain width W(D)=max over antichains A in P_D of sum_{T in A} 2^{-|union_{i in T} S_i|}. We conjecture that for every Boolean f with correlation eps with parities of degree <=a, the maximum linear bias of NW_{D,f} satisfies bias(NW_{D,f}) <= W(D) * eps, and moreover W(D) = Theta(m * 2^{-a}) on standard Galois-field NW designs, giving a tight Sperner-style replacement for the union-bound m * 2^{-a} factor in Nisan-Wigderson.

Rationale

The classical NW analysis pays a union-bound factor of m for the worst parity distinguisher, but the actual bias is a sum of Walsh coefficients indexed by subsets T of [m] whose contributions cancel along chains in the refinement poset; bounding cancellation by an LYM-type antichain width replaces the loose m factor with a combinatorially tight quantity. If the conjecture holds, NW seed length can shrink whenever W(D) « m, exposing a Sperner obstruction to natural-proofs-style attacks and giving a quantitative explanation for empirically observed sub-union-bound behavior.

Novelty

- Judge: `NOVEL` over 6 arXiv hits

Top hits consulted: - [1504.04131v2] Formulas for the Walsh coefficients of smooth functions and their application to bounds on the Walsh coefficients - [1905.12811v1] Embedding of Walsh Brownian Motion - [2601.03730v1] Perception-Aware Bias Detection for Query Suggestions - [1311.6531v3] Brains and pseudorandom generators - [1309.4930v2] The Zero-Undetected-Error Capacity Approaches the Sperner Capacity

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_74c5ddfd.py", line 153, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_74c5ddfd.py", line 121, in run_trial
    W_D = max(sum(2**(-len(S[i] for i in T)) for T in A) for A in combinations(range(1 << m), len(A)))
                                                                                              ^

NameError: name 'A' is not defined. Did you mean: 'a'?
```

Judge reasoning

The test crashed with a NameError before producing any data, so the pre-registered support condition cannot be evaluated. No (D,f) pairs were tested. | next: Fix the buggy line computing W_D (the generator references 'A' inside its own definition and misuses combinations); compute W(D) by enumerating antichains in P_D properly, then rerun the 300-pair sweep.

Kruskal-Katona Shadow Defect Lower-Bounds Monotone-KW Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Extremal set theory: the Kruskal-Katona shadow theorem and cascade representation of k-uniform antichains in the Boolean lattice (Kruskal 1963; Katona 1968; Frankl tradition). The 'KK shadow defect' delta_KK(A,k) := |partial(A)| - KK_lower(|A|,k) — slack between the actual (k-1)-shadow of a k-family A and its cascade-extremal lower bound — is a classical extremal-combinatorics invariant, computable in O(|A|*n) time, that has rarely been deployed against KW games (distinct from Bollobás set-pair inequalities used in monotone-circuit Razborov-approximation arguments). × Monotone Karchmer-Wigderson depth D_m(f) of monotone Boolean functions f:{0,1}^n -> {0,1}, equivalently monotone formula depth in the {AND,OR}-basis via the Karchmer-Wigderson equivalence; the KW relation is the bipartite search problem on (minterm, maxterm) pairs of f.
- **Recorded**: 2026-04-28 15:27 UTC
- **Entry ID**: `37788821c84c`

Statement

Let f:{0,1}^n -> {0,1} be monotone with minterm family M(f) sliced by Hamming weight as M(f) = bigsqcup_k M_k(f), and let partial(M_k) denote its (k-1)-shadow in 2[1]. Define the slice-wise shadow defect Delta_KK(f) := max_{1 <= k <= n} (|partial(M_k(f))| - KK_lower(|M_k(f)|, k)), where KK_lower(m,k) is the Kruskal-Katona cascade bound on the shadow of any k-uniform antichain of size m. Then for every monotone f on n <= 20 with |M(f)| >= 2, D_m(f) >= ceil(log_2(Delta_KK(f) + 2)).

Rationale

A monotone formula of depth d induces a balanced KW protocol whose 2^d leaves each fix a single coordinate witness for an entire surviving minterm-by-maxterm rectangle; this forces every weight-slice of M(f) to admit a covering by at most 2^d coordinate-stars, which in turn forces the slice to be near-compressed (low shadow defect) by a Frankl-style compression argument. KK-extremal slices correspond exactly to threshold-like minterm families which are known to admit O(log^2 n) monotone depth, while large defect at any slice is a structural irregularity invisible to the natural-proofs largeness condition (the property targets monotone depth only and is rare on random monotone clutters). The bound uses no relativization or algebrization, and stays inside the KW framework's safe directions.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1809.03967v2] More Cases Where the Kruskal-Katona Bound is Tight - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1801.08709v2] Adaptive Lower Bound for Testing Monotonicity on the Line

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a486861f.py", line 155, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
```

[1] n

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a486861f.py", line 112, in run_trial
    M_k = {compute_M_k(f_instance, k) for k in range(n + 1)}
           ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a486861f.py", line 42, in compute_M_k
    if sum(minterm) == k:
       ^^^^^^^^^^^^
TypeError: 'int' object is not iterable
```

Judge reasoning

The test crashed with a TypeError in compute_M_k before producing any data, so neither the support condition nor the falsification condition could be evaluated. | next: Fix the minterm representation bug in compute_M_k (iterate over bits of the integer-encoded minterm or pass tuple/bitmask consistently) and rerun the full enumeration on n=4,5 plus the n=6,7 sampled clutters.

Hodge Laplacian b_1 of Edge-Triangle Star-Complex Bounds SoS Degree of Tseitin

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Discrete combinatorial Hodge theory of finite simplicial 2-complexes (Eckmann 1944; Friedman 'Computing Betti Numbers via Combinatorial Laplacians'; Jiang–Lim–Yao 2011): $h_1(\Sigma)$ = nullity of the up-down Hodge Laplacian $L_1 = \partial_1^T \partial_1 + \partial_2 \partial_2^T$ over Q, equal to $\dim H_1(\Sigma;Q)$ by Hodge's theorem. Distinct from spectral graph theory (which uses only L_0), from Newton-polytope/lattice methods, and from zeta-function counts; rarely deployed against SoS proof complexity. × Sum-of-Squares (Lasserre/Parrilo) refutation degree $d_{SoS}(T(G,c))$ of unsatisfiable Tseitin instances on graphs G with odd total F_2 charge, accessed via the Schoenebeck/Grigoriev combinatorial proxy $g^*(F) = \min\{|T| : \oplus_{C\in T} a_C = 0 \in F_2^{V(G)}$ *and* $\oplus_{C\in T} b_C = 1 \in F_2\}$ — the canonical SoS refutation-degree target for binary CSPs (Schoenebeck 2008; Grigoriev 2001).
- **Recorded**: 2026-04-28 16:05 UTC
- **Entry ID**: `ea9308fd8a1c`

Statement

For a connected graph G with maximum degree Δ and odd-charge assignment c, build the simplicial 2-complex Σ(G) whose 0-cells are the edges of G, whose 1-cells are pairs of G-edges sharing a vertex, and whose 2-cells are the triples of G-edges meeting at a common vertex (the local 'vertex stars'). Let $h_1(G) := \text{nullity}(L_1)$ over Q. Conjecture: $g^*(T(G,c)) \geq \lceil h_1(G) / (\Delta\cdot(\Delta-1))\rceil + 1$ for every unsatisfiable Tseitin instance T(G,c) with $\Delta \leq 4$, and this bound is strict for at least one instance per girth-class of G.

Rationale

Schoenebeck-style SoS lower bounds for parity CSPs are governed by an F_2 dependency-girth that is sensitive to F_2 cancellation, while harmonic 1-cycles of the rationally lifted star complex Σ(G) detect independent topological 'holes' of the incidence structure that no single vertex-star 2-cell can fill — and these holes appear to obstruct precisely the local cancellation patterns short SoS pseudo-distributions exploit. Discrete Hodge theory is widely used in graph spectra and TDA but, to our knowledge, has not been deployed as a direct combinatorial obstruction inside the Lasserre/Parrilo hierarchy. If true, the conjecture yields a polynomial-time, basis-independent topological certificate of SoS hardness complementary to expansion-based bounds.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2602.03763v1] Optimizing Weighted Hodge Laplacian Flows on Simplicial Complexes - [1807.05044v5] Random Walks on Simplicial Complexes and the normalized Hodge 1-Laplacian - [2205.02134v1] Hodge Decomposition and General Laplacian Solvers for Embedded Simplicial Complexes - [1908.06789v2] Combinatorial Proof of the Minimal Excludant Theorem - [2204.12218v2] Combinatorial and Hodge Laplacians: Similarity and Difference

Empirical Test

- exit code: `124`, elapsed: `180.10s`

```
TIMEOUT after 180s
```

Judge reasoning

The test timed out after 180s without producing any RESULT line, so the pre-registered support condition could not be evaluated. Per Rule 2, a timeout yields INCONCLUSIVE. | next: Reduce trial count or graph sizes (e.g., n ≤ 12) and optimize the meet-in-the-middle SoS-degree search, or precompute h_1 separately, to fit within the time budget.

Continued-Fraction Spike of Truth-Table Rational Caps DNF_min

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Diophantine approximation via partial-quotient extremes: the Gauss-Kuzmin-Lévy distribution of continued-fraction coefficients and the rare-event tail of super-polylog partial quotients (Khinchin 1936; Bosma-Jager-Wiedijk; Hensley’s bounds). A classical analytic-number-theory invariant computed by one Euclidean-algorithm pass; distinct from Mahler measure / Lehmer heights and from spectral approaches, and rarely deployed in meta-complexity. × DNF_min(f) for f:{0,1}^n→{0,1} given by truth table — the Hirahara–Oliveira–Santhanam restricted-MCSP proxy in average-case-to-worst-case meta-complexity reductions, and its behavior under random p-restrictions (the Impagliazzo-Wigderson avg-to-worst hardness object).
- **Recorded**: 2026-04-28 16:32 UTC
- **Entry ID**: `6ead43796693`

Statement

For f:{0,1}^n→{0,1} let TT(f)∈{0,...,2^N−1} with N=2^n be its lex-ordered truth-table integer, α(f):=TT(f)/(2^N+1)∈[0,1)∩ℚ, and S(f):=max_{i≥1} a_i over the Euclidean continued-fraction expansion [a_0;a_1,...,a_L] of α(f). CONJECTURE: there exist absolute constants c_1,c_2>0 such that for every n≥4 and every f, if S(f) ≥ n^{c_1·log n} then DNF_min(f) ≤ 2^n·n / S(f)^{c_2}. Equivalently, a super-polylog Diophantine spike in α(f) is a one-sided structural certificate of polynomially-bounded DNF complexity, while the spike event has Gauss-Kuzmin probability < 1/poly(N) on random TTs.

Rationale

A coefficient a_i ≥ M in the CF of α(f)=p/q means p/q is exceptionally close to a rational p’/q’ of denominator q’ □ N/M, which forces the congruence TT(f)·q’ ≈ k(2^N+1) and makes TT(f) approximately periodic with period O(N/q’); approximate periodicity at period O(log q’) in N=2^n bits forces f to approximately depend on a junta of O(log q’) coordinates and hence to admit a small DNF. This realizes the Impagliazzo-Wigderson avg-to-worst paradigm structurally: outside the rare Gauss-Kuzmin spike tail, no Diophantine shortcut to low-DNF can occur, so DNF-hardness on the typical regime is the worst-case object. Because the implication is one-sided ‘rare signature ⇒ low complexity’ (not ‘large signature ⇒ high complexity’), the property is not large and sidesteps Razborov-Rudich naturalness.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1908.00121v4] Continued fractions over non-Euclidean imaginary quadratic rings - [2209.08318v3] Hausdorff dimension of sets with restricted, slowly growing partial quotients in semi-regular continued fractions - [1609.06986v1] More on Diophantine sextuples - [1603.03800v3] Diophantine approximation on matrices and Lie groups

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_338fbe98.py", line 145, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_338fbe98.py", line 78, in run_trial
    if dnf_complexity > 2**n * n / S_f**0.1:
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: '>' not supported between instances of 'NoneType' and 'float'
```

Judge reasoning

The test crashed with a TypeError (DNF_min returned None and was compared to a float), so no data was produced and the pre-registered support condition cannot be evaluated. Per rule 2, a crashed test yields INCONCLUSIVE. | next: Fix the DNF_min routine to always return an integer (or guard the comparison against None) and rerun the full pre-registered sweep over n∈{4,5,6,7} including the 5000 random TTs per n and adversarial m sweep.

Shifted Partials of Clause Cubic Lower-Bound Tree-Resolution Leaves

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Shifted partial-derivatives method in geometric complexity theory (Kayal–Saha–Tavenas 2016; Gupta–Kamath–Kayal–Saptharishi 2014): the dimension SPD_{□,k}(p)=dim_Q span{$x^{\beta\cdot\partial}$α p : |β|=□,|α|=k} of the multiplicative-shift of partial derivatives, a coordinate-free arithmetic-circuit lower-bound functional computable in O(n^6) by rank of an explicit Sylvester-like lift matrix. It strictly extends Mignon–Ressayre Hessian rank (= SPD_{0,1} up to constants) and is the dominant algebraic technique used to prove super-polynomial homogeneous depth-4 bounds in the Mulmuley–Sohoni GCT program; rarely deployed in proof complexity. × Tree-like resolution refutation leaf count L_T(F⊢⊥) of unsatisfiable random 3-CNF F at clause density α∈[4.5,5.5], equivalently lex-DPLL search-tree leaf count under unit propagation — the canonical proof-complexity proxy for refutation hardness.
- **Recorded**: 2026-04-28 16:59 UTC
- **Entry ID**: `3cb40990c2eb`

Statement

For an unsat 3-CNF F on n≤16 vars, form the clause cubic Q_F(x_0,…,x_n)=Σ_{C=(a,b,c)∈F} □_a·□_b·□_c where □_i^+=x_i, □_i^-=x_0−x_i, viewed in the monomial basis of Q[x_0,…,x_n]*3. Define ψ(F):=rank M*{1,1}(Q_F) − rank Cat_1(Q_F), where M_{1,1} is the (n+1)^2×(n+3 choose 3) coefficient matrix whose rows are the cubic coefficients of x_i·∂_j Q_F, and Cat_1 is the (n+1)×(n+2 choose 2) first-catalecticant. Conjecture: ψ(F) ≥ ⌊log_2 L_T(F⊢⊥)⌋ for every unsat 3-CNF F at clause density α∈[4.5,5.5]; a single instance with ψ(F)<⌊log_2 L_T⌋ falsifies it.

Rationale

Q_F algebraicizes the unsat witness as a degree-3 form; multiplying first partials by linear shifts probes the 'depth-4 expansion' that Gupta–Kamath–Kayal–Saptharishi exploited for the 2^√n IMM bound, exactly the GCT-aligned nonlinear information missed by the Hessian alone. Subtracting Cat_1 isolates the genuinely shifted contribution and steers clear of the prior Mignon–Ressayre attempt on the blacklist. Tree-resolution leaf count is a sharp proxy for DPLL hardness, so a quantitative ψ-vs-log L_T link would expose unsat-instance hardness as a Sylvester-rank obstruction.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity - [9906008v2] A Lower Bound on the Average-Case Complexity of Shellsort - [1801.08709v2] Adaptive Lower Bound for Testing Monotonicity on the Line - [2404.04038v1] Refutability as Recursive as Provability - [0804.4584v1] Feature Unification in TAG Derivation Trees

Empirical Test

- exit code: `124`, elapsed: `180.09s`

```
TIMEOUT after 180s
```

Judge reasoning

The test timed out after 180s (returncode=124) without producing any RESULT line or data, so the pre-registered support condition cannot be evaluated. No counterexample was observed either, so falsification is not established. | next: Reduce the workload (e.g., shrink to n∈{8,10,12} and 50 trials/cell, or precompute L_T with a faster DPLL) and raise the timeout so the rank computations and lex-DPLL counts can complete.

Hereditary Discrepancy of Clause-Variable Incidence Lower-Bounds Tree-Resolution Leaves

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial discrepancy theory: hereditary discrepancy herdisc(H) of a set system H, computed by the Lovasz-Spencer-Vesztergombi determinant lower bound detlb(H) = max_{k,S} |det(M)|^{1/k} over k x k submatrices M of the incidence matrix of H restricted to a subset S of columns, in the Chazelle-Matousek-Bourgain-Kashin discrepancy tradition (Matousek 'Geometric Discrepancy' Ch.4; LSV 1986). Rarely deployed against proof complexity; distinct from the Beck-Fiala slack approach and from spectral / hypercontractive bounds. × Tree-like resolution refutation leaf count L_T(F) of small unsatisfiable 3-CNF formulas F, equivalently lex-DPLL search-tree leaf count under unit propagation; the canonical proof-complexity proxy for refutation hardness.
- **Recorded**: 2026-04-28 17:25 UTC
- **Entry ID**: `15736a2ab8cf`

Statement

Let F be an unsatisfiable 3-CNF formula on n <= 20 variables and m clauses, and let H(F) be the clause-variable incidence hypergraph (clauses as hyperedges over n variables, ignoring sign). Let detlb(F) = max over 2 <= k <= min(8,m,n), and over column subsets S of size k drawn by greedy maximum-row-norm selection, of |det(M_S)|^{1/k} where M_S is the k x k submatrix of largest absolute determinant among k rows of the +-1 signed incidence matrix restricted to S (computed by exhaustive enumeration over the C(m,k) row choices, capped at 5000 random samples when

C(m,k) exceeds it). Conjecture: log2(L_T(F) + 1) >= c * detlb(F) * sqrt(n) for an absolute constant c >= 0.20, with a single counterexample (any F where the inequality fails) refuting the bound.

Rationale

The Lovasz-Spencer-Vesztergombi determinant lower bound is the strongest constructive lower bound on hereditary discrepancy and detects 'rich' submatrix structure that, in the Chazelle-Matousek-Bourgain-Kashin tradition, certifies dispersion-style obstructions; tree-resolution leaves on a CNF correspond to coverings of the input cube by partial-assignment cylinders constrained by clause incidence, so a high-determinant signed submatrix should force any DPLL branching tree to expose at least k linearly independent sign patterns and hence ~2^k leaves. Sqrt(n) calibration matches the typical Komlos / Spencer-style scaling for +-1 matrices and avoids the natural-proofs barrier because detlb is sensitive to clause signs (not just the underlying hypergraph) and is not closed under generic monotone shifts. The bridge is rarely traversed: Beck-Fiala slack has been tried (blacklisted) but determinant lower bounds and Chazelle's hereditary-discrepancy sampler have not been linked to DPLL leaves.

Novelty

- Judge: `NOVEL` over 11 arXiv hits

Top hits consulted: - [2212.02103v2] A Study of Hypergraph Using Null Spaces of the Incidence Matrix and its Transpose - [2111.02125v5] Expected Complexity of Persistence Barcode Computation via Matrix Reduction - [0702457v4] Topological invariants of piecewise hereditary algebras - [0906.0693v3] An improved lower bound on the counterfeit coins problem - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2ae6da60.py", line 136, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2ae6da60.py", line 116, in run_trial
    leaves = lex_dpll(F, 30e6)
             ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2ae6da60.py", line 31, in lex_dpll
    n = max(abs(var) for var, _ in F)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2ae6da60.py", line 31, in <genexpr>
    n = max(abs(var) for var, _ in F)
                         ^^^^^^
ValueError: too many values to unpack (expected 2)
```

Judge reasoning

The test crashed with a ValueError during clause unpacking in lex_dpll (clauses are 3-tuples, not 2-tuples), so no instances were evaluated and neither the support nor falsification condition can be assessed. | next: Fix the clause iteration in lex_dpll (unpack 3-literal clauses correctly, e.g., compute n via max(abs(lit) for clause in F for lit in clause)) and rerun the 200-instance battery.

Schur-Polynomial Positivity Defect Lower-Bounds Tree-Frege Lines for PHP

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Symmetric function theory: Schur-positivity of LR-style coefficients in the expansion of monomial symmetric functions associated to clause-occupancy vectors × Tree-like Frege (equivalently tree-resolution) refutation line count L_TF(F) on the pigeonhole CNF PHP_n^{n+1}
- **Recorded**: 2026-04-28 17:55 UTC
- **Entry ID**: `388ed224a992`

Statement

For each n in [3,9], let F=PHP_n^{n+1} and let v(F) in Z^n be the clause-occupancy vector v_i = number of clauses of F in which variable x_{i,} *appears (summed over holes), and form the monomial symmetric function m_{lambda(v)} where lambda(v) is the partition obtained by sorting v. Define the Schur-positivity defect S(F) = sum over partitions mu of n with mu different from lambda(v) of max(0, -<m_{lambda(v)}, s_mu>) where <,> denotes the inverse-Kostka pairing computed by Gaussian elimination on the Kostka matrix K_{mu,lambda}. Conjecture: log_2 L_TF(PHP_n^{n+1}) >= c* (S(F) + 1) for an absolute constant c >= 1/4, with equality up to a (1+o(1)) factor as n grows; one PHP_n^{n+1} with log_2 L_TF < (S(F)+1)/4 falsifies.

Rationale

Schur-positivity defects are sensitive to imbalance in symmetric clause-occupancy structure, and the PHP family's tree-resolution hardness is known to be driven by a symmetric matching obstruction; bridging Kostka inverses (a representation-theoretic invariant rarely seen in proof complexity) with PHP line counts may expose the symmetric kernel of EF lower bounds without invoking natural-proofs-style pseudorandom predicates. The Kostka matrix is integer, lower-triangular under dominance order, and invertible exactly by back-substitution, so the defect is a fully constructive number-theoretic statistic of the CNF, not a categorical invariant.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [0304041v1] $P \neq NP$, propositional proof complexity, and resolution lower bounds for the weak pigeonhole principle - [2511.20023v1] Lower Bounds for Bit Pigeonhole Principles in Bounded-Depth Resolution over Parities - [1611.08382v1] Conditionally positive definite kernels in Hilbert C^*-modules - [1501.05996v1] Double Kostka polynomials and Hall bimodule - [2404.04038v1] Refutability as Recursive as Provability

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f2496006.py", line 136, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f2496006.py", line 108, in run_trial
    L_TF = lex_dpll(F)
           ^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f2496006.py", line 77, in lex_dpll
    return backtrack()
           ^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f2496006.py", line 71, in backtrack
    if all(F[i][j] <= assignment[j] for j in range(n)):
```

```
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f2496006.py", line 71, in <genexpr>
    if all(F[i][j] <= assignment[j] for j in range(n)):
             ^
NameError: name 'i' is not defined. Did you mean: 'id'?
```

Judge reasoning

The test crashed with a NameError ('i' not defined) inside the lex_dpll backtrack function before any data was produced, so the pre-registered support condition could not be evaluated. | next: Fix the scoping bug in backtrack() (capture clause index i in the generator) and rerun the deterministic sweep over n in {3,…,9}.

Erdős-Rado Sunflower Petals Lower-Bound Lifted Log-Rank for IND_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Extremal set theory: the Erdős-Rado sunflower lemma — maximum number r(F) of sets in a family F admitting a common core Y with pairwise petal-disjoint complements S_i Y (Erdős-Rado 1960; Alweiss-Lovett-Wu-Zhang 2020 quasi-polynomial improvement; Rao 2020 simplification). Distinct from Razborov's monotone-circuit approximation method (which uses sunflowers on minterms in a min-AND-OR closure context), and rarely deployed as a quantitative invariant against deterministic communication complexity of lifted functions. × Deterministic two-party communication complexity of the indexing-lifted Boolean function f∘IND_2 for f:{0,1}^k → {0,1}, accessed via the real log-rank lower bound D^cc_log(f∘IND_2) := ⌈log₂ rk_R(M_{f∘IND_2})⌉ on the 2^k × 4^k lifted communication matrix M_{f∘IND_2}[x,y] = f(y_1[x_1], …, y_k[x_k]). This is the canonical Raz-McKenzie / Göös-Pitassi-Watson lifting target object at the smallest nontrivial gadget arity b=2.
- **Recorded**: 2026-04-28 18:31 UTC
- **Entry ID**: `f1f0ed613240`

Statement

For every Boolean function f:{0,1}^k → {0,1} with k ∈ {2,3,4}, let C_min(f) be the family of minimal 1-certificates of f (minimal partial assignments α ⊂ [k]×{0,1} with f|_α ≡ 1) and let r(f) be the maximum sunflower size in C_min(f), i.e. the largest m such that there exist α_1, …, α_m ∈ C_min(f) and a core Y ⊆ ⋂i *α_i with the petals α_i Y pairwise disjoint as literal sets. Then ⌊log₂ r(f)⌋ ≤ D^cc_log(f∘IND_2). Any f:{0,1}^k → {0,1} (k ≤ 4) with ⌊log₂ r(f)⌋ > ⌈log₂ rk_R(M*{f∘IND_2})⌉ refutes the conjecture.

Rationale

In the Raz-McKenzie / GPW lifting picture, IND_2 hands Bob a two-element 'choice' per coordinate while Alice picks the index, so a sunflower of m 1-certs with common core Y and disjoint petals should embed m linearly independent rank-1 patterns into M_{f∘IND_2}: the core fixes a shared structural part of the lifted matrix, while the disjoint petals act on disjoint y-coordinates and therefore should be R-linearly independent in the row span. Translating Erdős-Rado's structure-vs-randomness dichotomy into a log-rank witness would expose a combinatorial source of communication hardness orthogonal to spectral / discrepancy bounds. This bridge is also natural-proofs-safe: r(f) is not a 'large dense useful' property in the Razborov-Rudich sense (it is bounded by 2^k for k≤4 and tied to a specific algebraic invariant rk_R, not to arbitrary circuit size).

Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [1904.13056v4] Query-to-Communication Lifting Using Low-Discrepancy Gadgets - [2211.17211v2] On Disperser/Lifting Properties of the Index and Inner-Product Functions - [1809.10318v3] Improved Bound on Sets Including No Sunflower with Three Petals - [2505.03671v2] The Erdős-Rado Sunflower Problem for Vector Spaces - [2511.07739v3] A Lower Bound for the Fourier Entropy of Boolean Functions on the Biased Hypercube

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a37580e9.py", line 116, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a37580e9.py", line 92, in run_trial
    certificates = min_1_certificates(f, k)
                   ^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a37580e9.py", line 62, in min_1_certifi
    if f(alpha) == 1:
       ^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a37580e9.py", line 89, in <lambda>
   functions = [lambda alpha, f=f: all(f(alpha[:i] + (b,) + alpha[i+1:]) == 1 for b in [0, 1]) for alpha
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a37580e9.py", line 89, in <genexpr>
   functions = [lambda alpha, f=f: all(f(alpha[:i] + (b,) + alpha[i+1:]) == 1 for b in [0, 1]) for alpha
                                                ^
NameError: name 'i' is not defined. Did you mean: 'id'?
```

Judge reasoning

The test crashed with a NameError ('name i is not defined') in the lambda construction before any conjecture data was produced, so neither the support nor the falsification condition can be evaluated. | next: Fix the lambda scoping bug (e.g., bind i as a default argument in the generator) and rerun the full enumeration for k∈{2,3,4}.

Subword Complexity of Lifted Rows Lower-Bounds Real Rank for IND_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorics on words / symbolic dynamics: subword complexity p_w(n) = #{distinct contiguous length-n factors of w} and the Morse–Hedlund classification (Morse–Hedlund 1938; Cassaigne; Allouche–Shallit, 'Automatic Sequences' Ch.4). The Rauzy-graph factor-counting invariant is computable in O(|w|·n) by a single sliding-window pass. Distinct from Lempel–Ziv / RLE entropy (blacklisted) and from descent or inversion statistics; rarely deployed in lifting / communication complexity. × Deterministic two-party communication complexity of the indexing-lifted Boolean function f∘IND_2, accessed via the real-log-rank lower bound D^cc(f∘IND_2) ≥ ⌈log_2 rk_R(M)⌉ on the canonical 2^k × 4^k lifted communication matrix M[x,y] = f(y_1[x_1],…,y_k[x_k]) (Raz–McKenzie 1999; Göös–Pitassi–Watson 2015 lifting target at smallest nontrivial gadget arity b=2).
- **Recorded**: 2026-04-28 19:01 UTC
- **Entry ID**: `da476af6b27a`

Statement

For every Boolean function f:{0,1}^k→{0,1} with k ≤ 4, build the lifted matrix M = M_{f∘IND_2} ∈ {0,1}$^{\{2}$k × 4^k} and read each row r_x in lex order of y∈({0,1}$^{2)}$k as a binary string of length 4^k; let s(f) := max_{x∈{0,1}^k} p_{r_x}(k), where p_w(n) is the subword complexity. Then rk_R(M) ≥ s(f), so D^cc(f∘IND_2) ≥ ⌈log_2 s(f)⌉. A single (f,x) with p_{r_x}(k) > rk_R(M) refutes the conjecture.

Rationale

The IND_2 gadget injects a locally periodic alignment scaffold into every row of the lifted matrix, so two length-k windows of r_x that show distinct bit-patterns must witness distinct gadget-alignment cosets and we conjecture each contributes an independent direction to the column span. This pivots a classical Morse–Hedlund symbolic-dynamics invariant onto the lifted matrix in a way orthogonal to fooling-set, log-rank, and Möbius/Walsh attacks already tried. If true, subword complexity becomes a poly-time-computable, stable surrogate for partial log-rank that scales with gadget arity.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [0911.3482v5] Complexity of Networks (reprise) - [0101006v1] On Complexity and Emergence - [2401.14623v1] Structure in Communication Complexity and Constant-Cost Complexity Classes - [2508.11973v1] A robust family of residually finite groups; spectra of residual finiteness growth, computability properties, and other - [1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7cc2688d.py", line 94, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7cc2688d.py", line 42, in run_trial
    f = [f(x) for x in range(2**k)]
         ^^^^
TypeError: 'list' object is not callable
```

Judge reasoning

The test crashed with a TypeError ('list' object is not callable') due to a variable shadowing bug before any instances were evaluated, so no data was produced to assess the pre-registered support or falsification conditions. | next: Fix the name collision (rename the inner list, e.g., `truth_table = [f(x) for x in range(2**k)]`) and rerun the 300-instance sweep to obtain violation_count and mean tightness ratios.

Bakry-Émery Curvature Floor Lower-Bounds Tseitin Resolution Width

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Discrete Bakry-Émery Γ_2-curvature CD(K,∞) on finite graphs (Schmuckenschläger 1999; Lin–Yau 2010; Cushing–Liu–Münch 2019); the Bakry-Émery curvature κ_BE(v;∞) is the largest K such that Γ_2(f,f)(v) ≥ K·Γ(f,f)(v) for all f, computed locally as the smallest generalized eigenvalue of an explicit symmetric matrix Q_v built from the 2-neighborhood of v. A diffusion-/stochastic-analysis invariant distinct from Ollivier-Ricci (Skenderi 2025, blacklisted)

and Forman-Ricci (which collapses to a constant on 3-regular triangle-free graphs); rarely deployed in proof complexity. × Resolution refutation width $w(T(G,c) \vdash \perp)$ of unsatisfiable Tseitin formulas T(G,c) on 3-regular graphs G with odd total F_2-charge c, the Urquhart / Ben-Sasson–Wigderson / Alekhnovich–Razborov target. We use the lex-DPLL search-tree depth d_DPLL(T(G,c)) as a polynomially-tight proxy for w (Ben-Sasson–Wigderson).
- **Recorded**: 2026-04-28 19:42 UTC
- **Entry ID**: `6a507261ca55`

Statement

For every 3-regular graph G of girth ≥ 4 with odd-charge Tseitin instance T(G,c), the average Bakry-Émery curvature $\bar{\kappa}(G) := (1/|V|)\ \Sigma_v\ \kappa\_BE(v;\infty)$ is non-positive, and the resolution width obeys $w(T(G,c) \vdash \perp) \geq \lceil 0.4 \cdot |V(G)| \cdot (-\bar{\kappa}(G)) \rceil$. Equivalently, the rescaled invariant $\rho(G) := -\bar{\kappa}(G) \cdot |V(G)|$ is a Ben-Sasson–Wigderson-style width lower bound with multiplicative slack ≥ 0.4 across the cubic-girth-≥4 family. A single 3-regular girth-≥4 instance with $d\_DPLL < 0.4 \cdot |V| \cdot (-\bar{\kappa}(G))$ refutes the conjecture.

Rationale

Bakry-Émery curvature is a local Γ_2 obstruction quantifying how sharply diffusion concentrates near each vertex; on 3-regular triangle-free graphs it factors structurally through the 2-step walk pattern in a way that is provably finer than spectral gap (Cushing–Münch examples show $\bar{\kappa}$ separates graphs of equal λ_2). Tseitin width on cubic graphs is governed by edge-expansion (Ben-Sasson–Wigderson), and the discrete CD-Bonnet-Myers-type inequalities suggest $\bar{\kappa}$ controls a local 'mass-redistribution cost' that should match the bottleneck-counting argument behind Tseitin width. If true, this gives a coordinate-free local certificate for resolution width without invoking Cheeger or λ_2 explicitly.

Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [2103.10093v1] Conjectures on Convergence and Scalar Curvature - [1807.10181v2] Gradient estimates, Bakry-Emery Ricci curvature and ellipticity for unbounded graph Laplacians - [1608.07778v2] Bakry-Emery curvature and diameter bounds on graphs - [2403.15973v1] Isoperimetric profile function comparisons with Integral Ricci curvature bounds - [1809.07698v1] Discrete Ricci curvatures for directed networks

Empirical Test

- exit code: `124`, elapsed: `180.08s`

```
TIMEOUT after 180s
```

Judge reasoning

Test timed out after 180s (returncode 124) before producing any RESULT line or instance data, so neither the support nor falsification condition can be evaluated. Per the rules, a timeout yields INCONCLUSIVE. | next: Reduce the instance count or per-instance work (e.g., cap DPLL search depth, shrink to 24 instances, or precompute Bakry-Émery curvature) and rerun with a longer timeout to obtain the 72-instance metrics.

Dyck Zero-Crossing Count of Row-Walks Capped by Log-Rank

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Catalan / Dyck-path combinatorics — the zero-crossing count R(w) := #{t ∈ [N] : Σ_{s≤t} w_s = 0} of a ±1 walk w of length N (Feller's discrete arcsine / ballot statistic, Stanley EC2 Ch.6; Pitman 'Combinatorial Stochastic Processes' 2006; Aldous-Pitman Brownian-excursion scaling). The arcsine law fixes E[R(w)] ~ √(2N/π) for uniform random ±1 walks. One-pass O(N) per row via prefix-sum, distinct from Fourier-Walsh / RSK / descent / Eulerian / hypercontractive statistics and rarely deployed against communication complexity. × Real rank rk_R(M_f) of the 2^n × 2^n ±1 communication matrix M_f[a,b] = (-1)^{f(a,b)} of a Boolean function f:{0,1}^n × {0,1}^n → {0,1}, the canonical log-rank-bound target for deterministic two-party communication complexity D^cc(f) ≥ log_2 rk_R(M_f) (Mehlhorn-Schmidt 1982; Lovász log-rank conjecture; Lovett 2014).
- **Recorded**: 2026-04-28 20:08 UTC
- **Entry ID**: `eb402b6828f3`

Statement

For every Boolean f:{0,1}^n × {0,1}^n → {0,1} with N := 2^n and ±1 matrix M_f, define the row-walk zero-count R_i(f) := #{j ∈ [N] : Σ_{j'≤j} M_f[i,j'] = 0} and set R(f) := max_i R_i(f). Conjecture: for every f with n ≤ 5 we have R(f) ≤ 4·√(rk_R(M_f) · N · log_2 N), and symmetrically with rows replaced by columns. A single f for which max(R_row(f), R_col(f)) exceeds 4·√(rk · N · log_2 N) refutes the bound.

Rationale

Rank-r matrices factor as M = AB^T with A,B ∈ R^{N×r}, so each row is a fixed R-linear combination of r column patterns, forcing zero-crossings of its prefix-sum walk to inherit Khintchine-type concentration over r Rademacher contributions plus a √log N tail; the arcsine law sets the unconstrained baseline at √N. Tying R(f) directly to log-rank routes a classical Catalan-path statistic into the Lovász/log-rank tradition without spectral, Fourier, or approximate-rank machinery, exposing a one-pass rigidity proxy in a corner of CC dominated by Razborov-Sherstov pattern-matrix and γ_2 methods.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1403.8106v1] Recent advances on the log-rank conjecture in communication complexity - [2510.02583v2] The Log-Rank Conjecture: New Equivalent Formulations - [1111.5884v1] An additive combinatorics approach to the log-rank conjecture in communication complexity - [2202.10205v2] Partial Dyck paths with Air Pockets - [0709.0878v1] Pattern Avoiding Ballot Paths and Finite Operator Calculus

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_26a29e57.py", line 123, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_26a29e57.py", line 95, in run_trial
    rk = matrix_rank(M)
         ^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_26a29e57.py", line 31, in matrix_rank
    echelon_form = gaussian_elimination([row[:] for row in A])
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_26a29e57.py", line 21, in gaussian_elim
```

```
    A[i], A[max_row] = A[max_row], A[i]
                       ~^^^^^^^^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in the gaussian_elimination routine before any instances were evaluated, so the pre-registered support criterion (≥800 instances) could not be assessed. | next: Fix the matrix_rank/gaussian_elimination implementation (guard against empty pivot columns and handle the max_row index correctly) or replace it with numpy.linalg.matrix_rank, then rerun the full ≥800-instance trial.

Cyclotomic Norm Floor of Truth-Table Character Sum Caps ACC^0[m] Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic number theory of cyclotomic integers — the field norm N_{Q(ζ_m)/Q}: Z[ζ_m] → Z computed as the determinant of the multiplication-by-α regular representation on the basis {1,ζ_m,...,ζ_m^{φ(m)-1}} reduced modulo Φ_m, in the Stickelberger–Lehmer–Mahler tradition (rarely deployed in circuit complexity; distinct from Mahler-measure-of-truth-table polynomial which is blacklisted, since the input here is a sum of roots of unity ζ_m^{int(x) mod m} living natively in O_K rather than a root-of-an-integer-polynomial height). × ACC^0[m] circuit size s_{ACC[m]}(f) of Boolean functions f:{0,1}^n→{0,1}, equivalently the Beigel–Tarui SYM∘AND size, the canonical Williams-2011 algorithmic-method target for NEXP ⊄ ACC^0.
- **Recorded**: 2026-04-28 20:35 UTC
- **Entry ID**: `a13755e34e77`

Statement

Fix m≥2 and let ζ_m=e^{2πi/m}. For f:{0,1}^n→{0,1} define α_f = (1/2$^{n)\cdot\Sigma_\{x\in\{0,1\}}$n} f(x)·ζ_m^{int(x) mod m} ∈ Q(ζ_m). Then there exist absolute constants C,m_0 such that for every m≥m_0 with at least two distinct prime factors and every f computable by an ACC^0[m] circuit of size s≥2, the cyclotomic norm satisfies −log_2 |N_{Q(ζ_m)/Q}(α_f)| ≤ C·φ(m)·log_2 s; equivalently, |N(α_f)| ≥ s^{−C·φ(m)}. A single (f,s) pair with |N(α_f)| < s^{−100·φ(m)} refutes the conjecture.

Rationale

By Beigel–Tarui every ACC^0[m] function is a SYM of s ANDs, so α_f decomposes additively into s elementary character sums each with constant-norm contributions in O_K=Z[ζ_m]; sub-multiplicativity of the absolute norm under sums (via the Hadamard-style determinant bound on the regular representation) suggests the total norm cannot dip below s^{-O(φ(m))}. Random f have |α_f|≈2^{-n/2}, hence |N(α_f)|≈2^{-nφ(m)/2}, exponentially below the bound — exposing exactly the algebraic gap Williams' algorithmic method exploits between structured ACC computations and generic truth tables. The cyclotomic ring is the natural home for MOD_m structure, yet has not been used as a circuit-size obstruction.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1611.06783v2] Cyclotomic polynomials at roots of unity - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [2110.13086v2] Quantum Algorithms and Lower Bounds for Linear Regression with Norm Constraints - [2603.27466v1] English translation of Frobenius' and Stickelberger's “On the theory of elliptic functions” - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity?

Empirical Test

- exit code: 1, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d2ef101.py", line 166, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d2ef101.py", line 126, in run_trial
    norm = compute_cyclotomic_norm(alpha_f, m)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d2ef101.py", line 98, in compute_cyclo
    n = int(math.log2(len(truth_table)))
                          ^^^^^^^^^^^
NameError: name 'truth_table' is not defined
```

Judge reasoning

The test crashed with a NameError ('truth_table' is not defined) inside compute_cyclotomic_norm before any data was produced, so neither the support nor falsification conditions can be evaluated. | next: Fix the variable scoping bug in compute_cyclotomic_norm (pass the truth table explicitly) and rerun the pre-registered sweep over m∈{6,10,15} and n∈{8,10,12,14}.

Hilbert-Compression Floor on Property-A Gadgets Forces α·Q(f) Multiplicity in Protocol Pullbacks

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Lifting (CGL); coarse geometry / large-scale topology of metric spaces (Roe algebras, Property A, Hilbert-space compression of finite-scale lifts) × Deterministic two-party communication complexity of lifted boolean functions f∘G^n, expressed as the minimum-multiplicity log of a RoeCover over the LiftedInputSpace at scale R_Π
- **Recorded**: 2026-04-28 23:06 UTC
- **Entry ID**: `15542d600c45`

Statement

Let G be a MetricGadget for which PropertyAExpand applied to a FolnerWitness produces a certified ε-Hilbert-space embedding with compression exponent α := h(G) > 0 (i.e., the embedding φ satisfies $||\varphi(u)-\varphi(v)||_2 \geq c\cdot d(u,v)^\alpha - C$ for fixed c,C). Then for every boolean predicate f : {0,1}^n → {0,1} with deterministic decision-tree query complexity Q(f), every deterministic communication protocol Π for f∘G^n yields a CoarsePullbackProtocol whose multiplicity m_Π at scale R_Π = diam(G) satisfies log_2(m_Π) ≥ α · Q(f). Combined with axiom A2 this gives CLC(f, G) ≥ α · Q(f), tightening axiom A4 in regimes where asdim_R(G) is hard to certify but a Følner witness is available.

Rationale

This is a direct stress-test of axiom A4 (Property-A Transfer) decoupled from axiom A3 (Asdim Amplification). The coarse embedding produced by PropertyAExpand sends any finite-multiplicity scale-R cover of (X×Y)^n into an ℓ^2 configuration whose ball-packing number scales like 2^{α·n}. A query-adversary argument on f then pushes the packing requirement up to 2^{α·Q(f)}, which by axiom A2 lower-bounds protocol cost. The conjecture should hold because Property A alone — without quantitative asdim bounds — already supplies the metric distortion needed to beat trivial covering arguments, and it is the minimal coarse hypothesis under which Yu-style descent is known to apply.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1411.4413v2] Observation of the rare $B_s^0 \to \mu^+\mu^-$ decay from the combined analysis of CMS and LHCb data - [0901.0512v4] Expected Performance of the ATLAS Experiment - Detector, Trigger and Physics - [2601.07595v3] Deep Search for Joint Sources of Gravitational Waves and High-Energy Neutrinos with IceCube During the Third Observing R - [2312.08907v2] Roe algebras of coarse spaces via coarse geometric modules - [2408.07701v2] A categorical interpretation of Morita equivalence for dynamical von Neumann algebras

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5de7cf2b.py", line 132, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5de7cf2b.py", line 89, in run_trial
    φ = property_a_expand(X, Y, F)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5de7cf2b.py", line 47, in property_a_ex
    L = [[d_pluss(X[i], X[j]) for j in range(len(X))] for i in range(len(X))]
          ^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5de7cf2b.py", line 46, in <lambda>
    d_pluss = lambda x, y: sum(hamming_distance(xi, yi) for xi, yi in zip(x, y))
                           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5de7cf2b.py", line 46, in <genexpr>
    d_pluss = lambda x, y: sum(hamming_distance(xi, yi) for xi, yi in zip(x, y))
                               ^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5de7cf2b.py", line 18, in hamming_dista
    return sum(xi != yi for xi, yi in zip(x, y))
                                      ^^^^^^^^^
TypeError: 'int' object is not iterable
```

Judge reasoning

The test crashed with a TypeError in hamming_distance before any trial completed, so no data was produced to evaluate the pre-registered support or falsification conditions. | next: Fix the d_pluss/hamming_distance type handling (ensure gadget elements are iterable bitstrings, not ints) and rerun the 45-trial battery.

Effective-Resistance Diameter Lower-Bounds Tseitin Resolution Width

- **Verdict**: INCONCLUSIVE
- **Bridge**: Electrical network theory on graphs (Kirchhoff effective resistance and the Doyle-Snell / Lyons-Peres random-walk metric) × Resolution refutation width w(T(G,c)⊢⊥) of Tseitin formulas on 3-regular graphs G with odd total F_2-charge c, accessed via lex-DPLL search-tree depth d_DPLL(T(G,c)) as a polynomially-tight Ben-Sasson–Wigderson proxy
- **Recorded**: 2026-04-28 23:09 UTC
- **Entry ID**: 23fb9df5b1a2

Statement

Let G be a connected 3-regular graph on n≤40 vertices with odd F_2-charge c, and define the resistance-diameter invariant ν_R(G) = min_{S⊂V, |S|=⌈n/3⌉} max_{u∉S} R_eff(u,S), where R_eff(u,S) is the effective resistance between vertex u and the set S in the unit-conductance electrical network on G (computed by one pseudoinverse of the graph Laplacian). We conjecture there exist absolute constants c_1>0, c_2>0 such that d_DPLL(T(G,c)) ≥ c_1·ν_R(G)·n − c_2 for every odd-charge c, and moreover ν_R(G) = O(1/n) on every family of non-expanders (graphs of bounded Cheeger constant→0), so the bound is automatically vacuous off the expander regime. A single (G,c,n≤40) instance with d_DPLL < c_1·ν_R·n − c_2 for the empirically fitted (c_1,c_2) refutes the conjecture.

Rationale

Effective resistance is the canonical electrical avatar of expansion (Chandra-Raghavan-Ruzzo-Smolensky-Tiwari; Lyons-Peres) and tightly tracks the Cheeger / spectral-gap scale that Ben-Sasson–Wigderson tied to Tseitin width, yet R_eff has not been used directly as a Tseitin-resolution invariant — prior expander-based bounds use λ_2 or edge expansion, not the resistance-diameter against a balanced set. The resistance-diameter ν_R is a global geometric statistic (a 'random-walk diameter') that should distinguish expanders (ν_R = Θ(1/n)·something_big after rescaling vs locality) from path-like graphs where R_eff scales linearly and DPLL is short, giving a falsifier-friendly quantitative bridge. If true, ν_R supplies a one-Laplacian-pseudoinverse certificate of Tseitin hardness, refining the spectral-gap heuristic with a vertex-resolved invariant.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations - [2101.11818v1] Contagion-Preserving Network Sparsifiers: Exploring Epidemic Edge Importance Utilizing Effective Resistance - [0407622v3] A Study of the Composition of Ultra High Energy Cosmic Rays Using the High Resolution Fly's Eye - [1903.12243v1] DEEP-FRI: Sampling outside the box improves soundness - [1111.5884v1] An additive combinatorics approach to the log-rank conjecture in communication complexity

Empirical Test

- exit code: `1`, elapsed: `0.17s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e44f1ca6.py", line 127, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e44f1ca6.py", line 94, in run_trial
    nu_R = min(max(effective_resistance(G, S[:i]) for i in range(1, len(S)+1)), max(effective_resistan
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e44f1ca6.py", line 94, in <genexpr>
    nu_R = min(max(effective_resistance(G, S[:i]) for i in range(1, len(S)+1)), max(effective_resistan
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e44f1ca6.py", line 65, in effective_res
    L_inv = pseudoinverse(L)
            ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e44f1ca6.py", line 54, in pseudoinverse
    return [[L_inv[i][n+j] for j in range(n)] for i in range(n)]
             ~~~~~~~~~^^^^^
TypeError: 'Fraction' object is not subscriptable
```

Judge reasoning

The test crashed with a TypeError in pseudoinverse (Fraction object not subscriptable) before any data was collected, so the pre-registered support condition could not be evaluated. | next: Fix the pseudoinverse implementation (return a proper 2D matrix structure) and rerun the full 240-instance protocol before drawing conclusions.